Not Finished, But Abandoned

by

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ACKNOWLEDGMENTS

It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....

Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.

— David C. Makinson (1965)

Above is the famous "preface paradox," which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

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ABSTRACT

FIXME: basically a placeholder; do not believe

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,

2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

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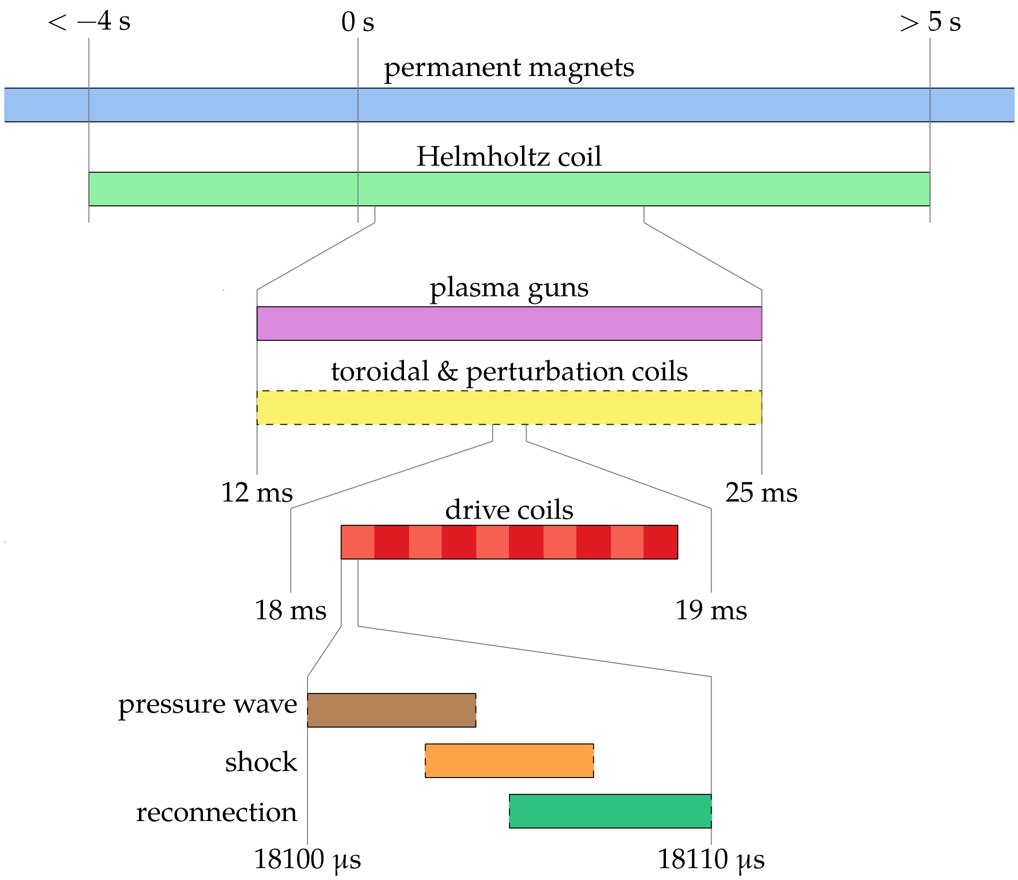

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3 MULTI-TIP LANGMUIR PROBE

3.1 What's a Langmuir probe

Hershkowitz: Hershkowitz (1989) Hutchinson: Hutchinson (2002b) Olson: Olson (2020)

Plasma diagnostics make a broad phylogenetic tree ranging from the variety of plasma parameters they can measure to their size and complexity. In the Big Red Ball we prefer simplicity and cheapness to some of the wilder diagnostic varieties so a common probe that we use is a Langmuir probe.

3.1.1 Basic Construction

These Langmuir probes are quite basic in construction. At it's simplest, a conductor is placed in contact with the plasma and a voltage is applied relative to ground which we will call the probe voltage, V_{probe}. This will locally perturb the plasma and collect a current, I_{probe}, which can be seen in figure 3.1. If $V_{probe} < V_{plasma}$, an electric field is formed that repells electrons and attracts ions and the inverse for $V_{probe} > V_{plasma}$.

Langmuir probes have been developed to work in a very wide range

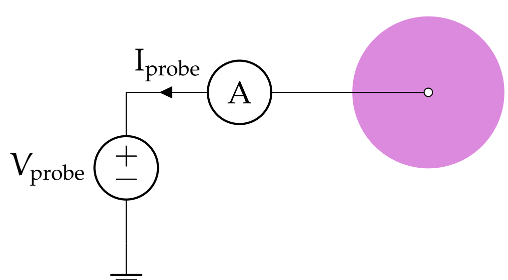


Figure 3.1: A circuit for a very simple Langmuir probe. The probe is biased to the probe voltage, V_{probe}, relative to ground by the power supply. The conductor (white circle) collects ions and electrons from the plasma (pink circle) causing a current to the probe, I_{probe}.

of plasma parameter configurations that well encompass the plasma parameters TREX achieves (see table 1.2).

3.1.2 Extracting Plasma Parameters

The most challenging part of these probes is the conversion of measured current to plasma parameters. Since we are trying to get a local measurement of the plasma parameters and we are also perturbing the plasma locally by touching it with a probe, we need to create a theory that correctly takes into account that perturbation such that we can get the plasma parameters. This often consists of a fitting procedure where the difference between the theoretical and experimental I-V curves is minimized.

Additionally, many a probe theory is purpose built for plasmas with no magnetic field. As we are studying magnetic reconnection in which there are in fact magnetic fields, this could change the I-V characteristic in unexpected ways. We'll discuss below why these effects are not very impactful and in future sections describe the theories we use for extracting the plasma parameters.

3.1.3 Plasma Perturbation

A problem often found in highly magnetized plasma devices is the long range perturbation to the plasma by the probe. In the Big Red Ball there are times where we have a relatively strong field and times when we don't so let's examine both cases.

3.1.3.1 Unmagnetized Perturbation

Let's examine this from an unmagnetized starting point. In this configuration, the probe collects nearly isotropically to an extent limited by the Debye length which creates a sheath around the probe and operates to shield the electric field caused by the difference between V_{probe} and V_{plasma}.

This length scale is very short and plasma can easily refill the sheath in this unmagnetized system.

3.1.3.2 Magnetic Field Effects

Now in a highly magnetized system (gyroradius far smaller than probe length scale) the plasma interacting with the probe comes from a flux tube approximately the size of the probe area projected into the magnetic field direction. Because of this, the probe can start "emptying out" the flux tube because of the difficulty of plasma to diffuse across field lines into the flux tube. This means the extend of the probe perturbation is far longer.

3.1.3.3 Perturbations in our System

We are quite lucky in that our system geometry reduces the impacts of flux tube emptying and the plasmas we are studying are often relatively unmagnetized.

3.1.3.3.1 Geometric Considerations The TREX experiment is (often) cylindrically symmetric and our probes are placed in different toroidal planes (see figure 2.6 for coordinate system reference). This means that without any toroidal field, any emptying only occurs in the plane of the probe which may still affect the probe I-V characteristic but is not affecting other probes measurements in a significant manner. When we do have a toroidal field, we also often have non-uniform B_r and B_z which means the flux tube passes through many different plasma parameters. This may help to average out any flux tube emptying because at different points along the flux tube more plasma may be able to enter the tube than just what is expected at the probe causing a reduced perturbation length.

3.1.3.3.2 Magnetic Field Considerations As we have magnetic fields in our system, we do need to worry about these flux tube effects. Fortunately, our magnetic field are relatively small and in the region we are often most interested in (where reconnection is occurring) the magnetic field is far smaller than the initial background field. We characterize this smallness by the ratio of the gyroradius to the radius of the probe area which is on the order of ??? (see 3.1).

3.2 Hardware

There has been a long and storied history of probe building on the Big Red Ball expecially for the development of multi-tip Langmuir probes. For some more information about our probes, see Olson (2020).

3.2.1 Physical Constraints

Our Langmuir probes need to satisfy a number of requirements.

- **Length**: The probe is attached to a port on the surface of the machine. This means it starts at a spherical radius of 1.5 m and must be inserted from there. Thus the insertable length should be fairly long and must be stable. This is described in 3.2.2.

- **Material**: We want our probe to survive the plasma exposure and not affect our vacuum pressure via offgassing. We also don't want to have to worry about secondary electron emission. We discuss this in 3.2.3.

- **Probe Size**: We want to measure the plasma locally so our probe body needs to be small relative to the plasma gradients. It also needs to be able to fit through ball joints that can be mounted to the machine that have a radius of ???. These considerations are taken into account in 3.2.3.

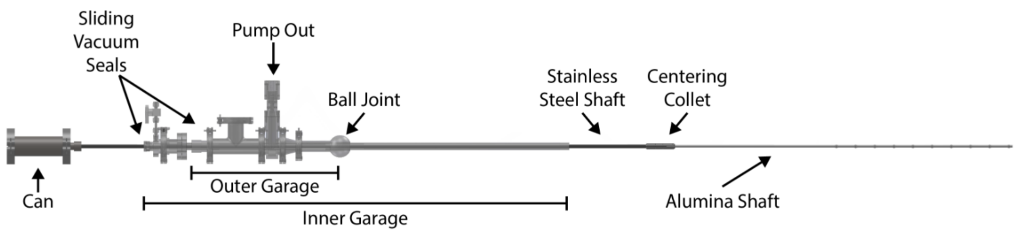


Figure 3.2: Three stage probe shaft with an outer and inner garage and the main stainless steel shaft that connects to the probe.

3.2.2 Shaft Components

For our probe to reach everywhere we want and also actually be in the position we desire we need a long shaft with a sufficient amount of support to be accurate to a few centimeters. Why a few centimeters? Our plasma dynamics are reproducible but not to such an extreme that a centimeter is noticeably different. Also it's difficult with the probe length to achieve too much better.

3.2.2.1 Garages

As can be seen in figure 3.2, the probe shaft is a multistage system. Each stage consists of some tubing where connections between different stages are made using quick-connect high-vacuum fittings (like part 3486N32 from McMaster-Carr).

We mount the probe to the machine using a flange on the outer garage or a ball joint (as seen in the figure). Then we can move each garage independently to insert the probe a desired length.

3.2.2.2 Shaft materials

We need to sufficiently support our probe and the shafts themselves so we use stainless steel until we enter the plasma region in which we change to alumina.

For the shafts, stainless steel is vacuum safe, tolerant to plasma interaction, and can support the weight of the structure easily without excessive flexing. Unfortunately, as a conductor, it will act as a long Langmuir probe that will create a ray in the machine at a single potential. This will perturb the plasma much more significantly than using an insulator.

To that end we use alumina, a high temperature ceramic who's surface will accumulate charge such that it'll be left at the floating potential (the potential at which the ion and electron currents balance). This helps to perturb the plasma less as there is no net current to the shaft and thus the plasma should (hopefully) be less perturbed.

Right near the end of the probe shaft, we use stainless steel tubing to connect from the alumina to the probe body. This is to reduce the overall size of the shaft in perturbing the plasma at the probe and is only for 10 cm of length.

3.2.2.3 Centering Collet

Each shaft/garage, is held in place by the sliding vacuum seal that connects it to it's larger garage. Unfortunately this gives only a single point of contact for making the shafts concentric. This would cause excessive drooping of the probe and may be so bad as to cause the probe to hit the edge of the machine when being inserted/removed.

To fix this issue, we have used centering collets that allow the inner shaft/garage to move while being held in place by the outer garage. These can be both machined or 3D printed. When 3D printed we can make these to also hold the o-rings so they are double-acting.

3.2.2.4 Feed-thru

We need some way to extract the signals from the probe tip. Because of the construction of our probe and for simplicity purposes, the probe shaft is at vacuum pressure. Thus we attach a feed-thru to the end of the can

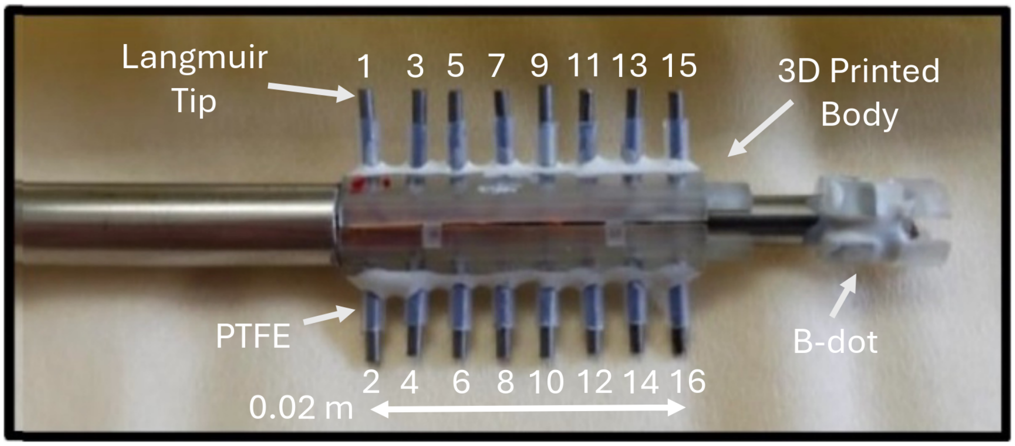


Figure 3.3: Picture of the end of the multi-tip Langmuir probe. Image from Gradney (2025).

from figure 3.2. This is constructed using two DB-25 hermetically sealed connectors where we solder the wires directly to the vacuum side of the can and then can easily move and connect the probe externally.

3.2.3 Tip Components

3.2.3.1 Probe body

Our probe does not need to withstand extreme heat fluxes and our vacuum conditions don't need to be perfect so we are lucky enough to be able to use a 3D printed probe body.

This is constructed using stereolithography which gives a sufficient resolution for the probe. Additionally the plastic used holds up well in vacuum. The body must connect to two different diameter stainless steel tubes (as can be seen in figure 3.3). Thus we use two annuli whose inner/outer radii match the tubing.

The tubing exists to connect the probe to the main shaft holding it up and also a small B-dot coil which is discussed in 3.2.4.

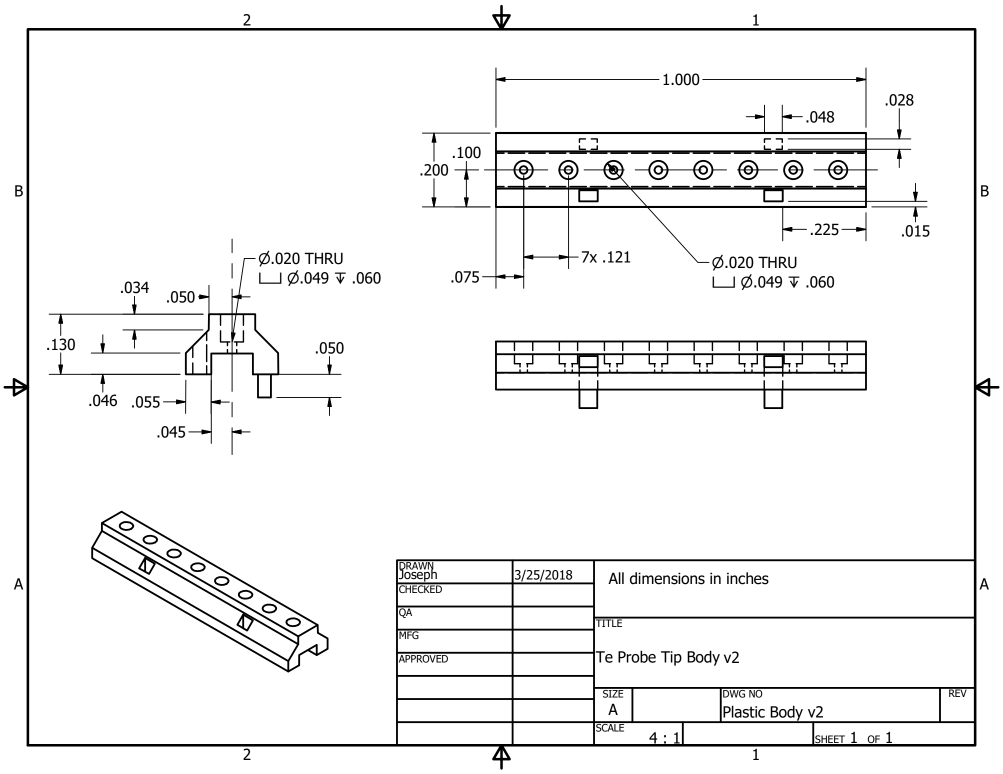


Figure 3.4: Probe body from engineering sketch. Created by Joe Olson.

3.2.3.2 Tip material + size

To collect current, we use 16 separate tips which each sample a point on the IV curve. We make these tips out of Molybdenum which requires us to connect the tips to the signal wire using a friction fit.

3.2.3.2.1 Why use Mo Secondary electron emission from molybdenum is quite small as compared to many metals Dekker (1958) which helps with IV-characteristic interpretation Hershkowitz (1989). Additionally, as a refractory metal, molybdenum will not melt nor sputter at the high temperatures the probe tips may get to as they collect current.

3.2.3.2.2 Size considerations We expect the current to the tips to be on the order of 1 kA/m^2. We measure the current across a 'sense' resistor (as discussed thoroughly in 3.3.2) but we don't want to have to use high power resistors for this. Thus we choose our current to be on the order of 10 mA which is feasible by having a tip area on the order of 10 mm^2. This amount of current also gives a reasonable signal-to-noise ratio.

Unfortunately, the probe is not all that cylindrical which means that the idealized probe theories we use are not as precise as we may have hoped. Future probe designs could use longer and smaller diameter tips to avoid this issue.

3.2.3.3 Tip Shielding

We protect each tip from the base of the probe body using PTFE/teflon. Teflon isn't great in a vacuum environment because of offgassing but does the trick and is cheap and easy to use. After inserting the tip into the probe body we push the teflon on and apply a small amount of vacuum-safe epoxy at the base to hold the shielding in place.

3.2.3.3.1 Why shield at all? We could potentially shorten each tip so the collecting area extends all the way to the probe body. Unfortunately because of the increased size of the body this can cause shadowing effects where tips on one side of the probe have a different IV-characteristic than those on the other. This 'shadowing' is caused by plasma flow parallel to the tip direction whcih gets intercepted by the probe body.

3.2.3.3.2 First order mach effects to avoid Another reason to shield the probe is to avoid mach effects that would cause different IV-characteristics for tips on opposite sides. As can be seen in figure 3.5, in an ideal mach probe the ions are prevented from easily accessing the "shadowed" side of the probe (right side) because of their inertia. If we shrink the probe

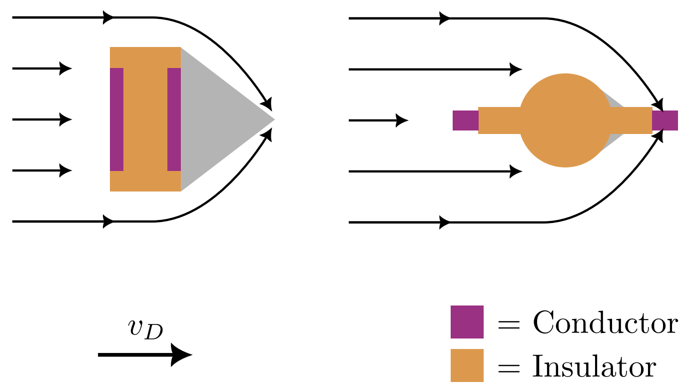


Figure 3.5: Simplified cartoon of the expected Mach effect for the multitip probe. The black arrows in the top figures are the approximate paths ions take due to the electric fields around the probe. The grey shaded region is where the ion density is significantly diminished. The left probe is a classic Mach probe geometry and the right probe is the multitip probe whose geometry is purpose built to avoid this Mach effect. Figure inspired by Merlino and D'Angelo (1987), Fig. 10.

body and increase the distance the tip is from the probe the mach effect is reduced because ion inertia is not sufficient to prevent collection. This is only approximate though and we sometimes do see effects that could be caused by shadowing. For an in depth discussion of mach probes we refer the interested reader to Chung (2012).

3.2.4 B-dot Coils

Since we don't know the exact location of the probe and the plasma is not perfectly reproducable, it's especially useful to have measurements we can compare to other diagnostics to locate the probe in space and around the reconnection region. To do this we use hand wound three axis Bdot coils that can measure the change in magnetic field near the tips.

3.2.4.1 Size Considerations

Through past trial and error and many a day spent hand winding coils for past students, we've found a reasonable vector area magnitude to use for the B-dot coils. During reconnection the largest $\partial_t \mathbf{B}$ is on the order of 10 kT/s. Thus, from Faraday's law,

$$\mathcal{E} = \mathbf{A} \cdot \partial_t \mathbf{B} \tag{3.1}$$

where $\mathcal{E}$ is the electromotive force and $\mathbf{A}$ is the vector area of the coil Jackson (1999). This means that for a signal of 1 V we are looking for a cross-sectional area of 10^{-4} m^2.

Using a 3D-printed B-dot body (seen in right in figure 3.3) we wrap wire around a square shaped center six times giving an area of $\sim 5 \times 10^{-5}$ m^2. Although smaller than that required for a 1 V measurement, this makes sure the signal doesn't become too large should some kind of resonance exist in the circuit.

3.2.4.2 Noise Reduction

Since common mode noise can relatively easily couple to the coils, we use twisted pair for each wrapping direction. This gives us two circuits that have nearly opposite vector area, $\mathbf{A}_1 \approx -\mathbf{A}_2$, such that we can subtract the two measurements and thus remove common mode contributions. We will discuss this more in 3.3.3.4.

3.2.5 Length Scales

Table 3.1 contains all information that helps us decide on our Langmuir probe theory and signal interpretation. There are a few key features to note that let us validate the use of finite area B-dot coils and multiple tips.

Name	Symbol	Equation	Value
Tip radius	r	—	5e-4
Exposed tip length	l	—	2e-3
Neighboring tip separation	—	—	2e-3
Across probe tip separation	—	—	1.5e-2
B-dot coil side length	—	—	3e-3
Debye length	λ_D	$\sqrt{\frac{\epsilon_0 T_e}{ne}}$	3e-5
Ion Gyroradius	r_i	$\frac{m_i\sqrt{2m_i\bar{v}_i}}{eB}$	6e-1
Electron Gyroradius	r_e	$\frac{m_e\sqrt{2m_e\bar{v}_e}}{eB}$	1.5e-1
Ion mean free path	λ_i	$\frac{\bar{v}_i}{\nu_{i,i}}$	9e-4
Electron mean free path	λ_e	$\frac{\bar{v}_e}{\nu_{e,e}}$	4.5
Electron energy relaxation length	λ_ε	* Eq. 5	4.5
Sheath thickness	h	$\sim \lambda_D$	3e-5

Table 3.1: Length scales of the multitip probe. All lengths are in meters and are approximate. For the interested reader and for the energy relaxation length, please refer to Demidov et al. (2002) for Langmuir probe relevant physics of these scales. These parameters were calculated with $n = 10^{18}$ m^{-3}, $T_e = 20$ eV, $T_i = 0.1 \times T_e$, and $B = 10^{-4}$ T. For the mean free paths we have chosen the frequency that gives the shortest length.

Our ion and electron gyroradii are both relatively large compared to the probe tips and B-dot coils. This means that if we assume the plasma is relatively uniform in a flux tube, the entire probe is sampling a quite uniform plasma.

3.3 Circuitry

The multitip Langmuir probe consists of a number of circuits that work together to give us some nice measurements. Here we enumerate their operation methods, upgrades, signal extraction, and calibration techniques.

3.3.1 Bias Source

We have thus far had two different circuits that we've used for biasing each tip relative to the laboratory ground. The first, made by Jan Egedal while at VTF is manually controlled, while the second, made by me is remotely operated. We'll discuss how they work and their pros and cons.

3.3.1.1 Manually Controlled

Our previous setup consisted of multiple electrolytic capacitors who's voltage is set by a series of resistors that split the voltage between the negative power supply (V_-), ground, and the postivie power supply (V_+).

3.3.1.1.1 Principles of operation Figure 3.6 shows a 4 tip biasing circuit (our circuit uses 16 tips but we have simplified it). The 'rails' (which are the maximum and minimum voltage of the circuit) are set by power sources at V_+ and V_- which are biased relative to the ground input connection labeled `Jgnd`. A series of resistors splits the voltage between the rails and we choose one of the tips to be at laboratory ground. This lets us set the maximum and minimum voltage along with setting how many tips are positively/negatively biased. Also, moving the ground changes the voltage separation between tips so we can get higher resolution. The voltage separation charges the respective capacitors for each tip to a bias voltage, $V_b =$ `Vb`. The capacitors exist to keep the bias voltage of each tip nearly constant during an experimental shot.

Now let's calculate the bias for each tip. Let $i, j \in \{1, \ldots, N\}$ be the index of the tip and the index of the ground connection respectively. Then,

$$V_{bi} = \begin{cases} \frac{V_+}{j}(j - i) & \text{if } j \geqslant i \\ \frac{V_-}{N+1-j}(i - j) & \text{if } j < i \end{cases} \tag{3.2}$$

where V_{bi} is the bias voltage for tip i.

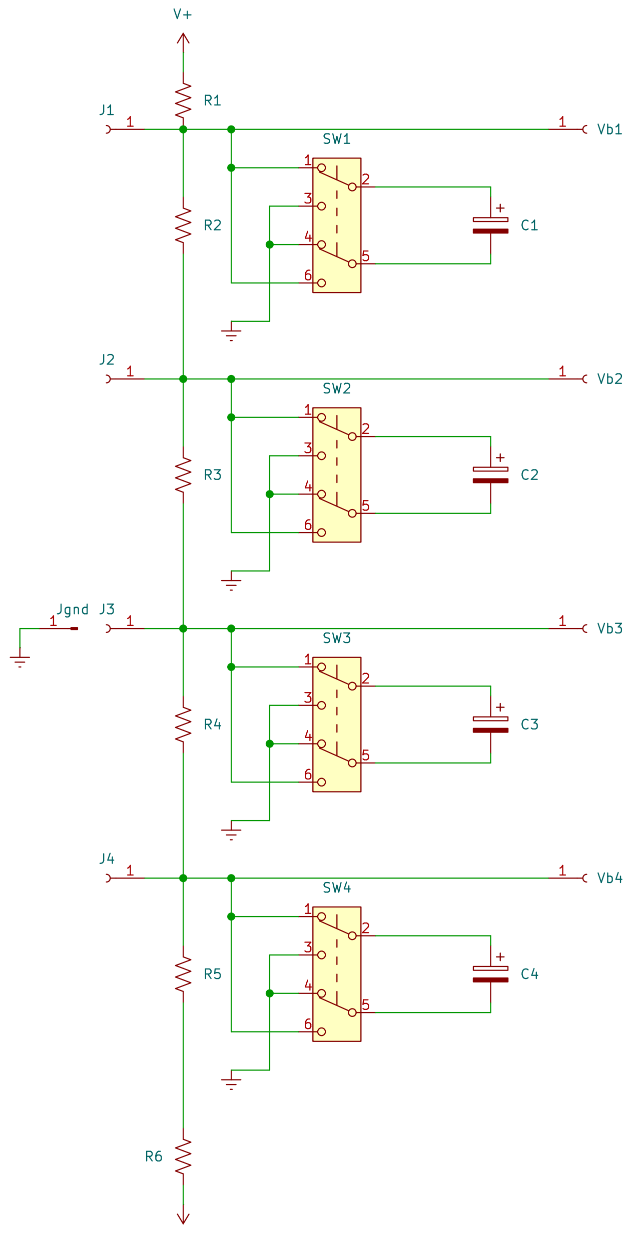


Figure 3.6: Old biasing circuit for the multitip Langmuir probe.

3.3.1.1.2 The good, the bad, and the ugly Because of the simplicity of this system, it has proven to work relatively well since it's been built. Unfortunately since most of it's components are free-standing and the wired connections undergo some movement during travel and assembly, various solder joints are prone to breaking. Additionally because of the need to manually flip the polarity of each electrolytic capacitor using a switch mounted to a board and also move the ground via a banana connection, it is infeasible to remotely operate the bias supply. Additionally this circuit has no protection against setting the tip voltages extremely far apart which may cause arcing between wires or tips.

3.3.1.2 Remote Controlled

We've upgraded the previous setup by allowing the bias to be remotely controlled. This is done by removing the switches needed in the past setup and instead having one power supply set the voltage of the most negative tip and the other power supply set the maximum bias voltage difference between tips.

3.3.1.2.1 Principles of operation This system operates by having a large capacitor that is set to some negative voltage relative to ground. Then, all the other individual capacitors for each tip has it's negative terminal set to the output of the large capacitor. Thus we can change the bias of all the tips by the same amount by changing the negative bias. The splitting of the bias voltage is similar to the previous setup where the voltage from the positive power supply is split using a series of resistors. The circuit schematic and images of the circuit can be seen in figures 3.7 and 3.8 respectively.

We can write out the individual tip biases mathematically as

$$V_{bi} = \frac{V_+}{N+1} i + V_- \tag{3.3}$$

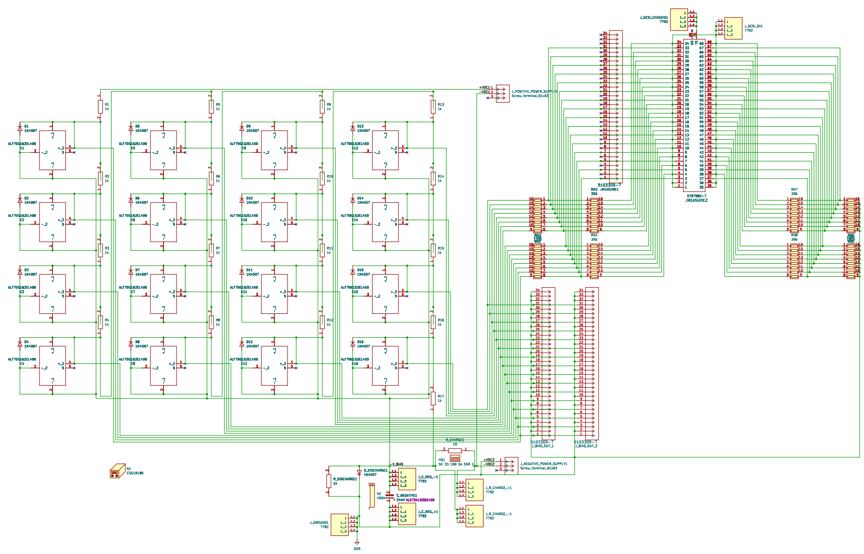
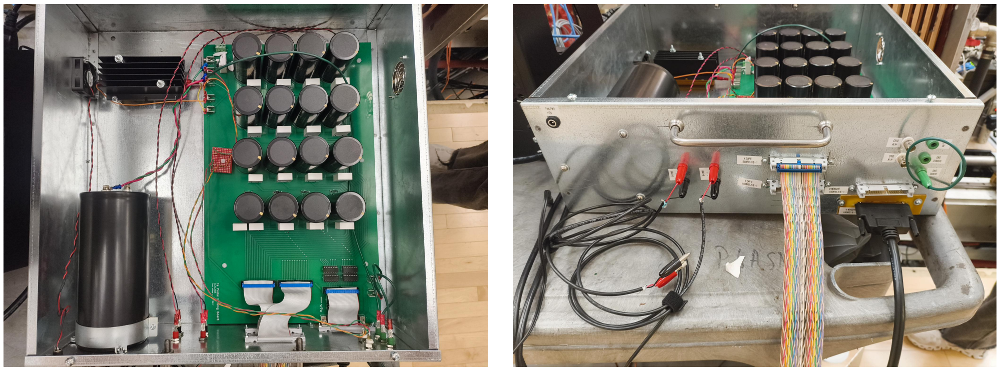


Figure 3.7: Circuit diagram of the new biasing setup. The left side consists of capacitors with reverse voltage protection diodes. Each capacitor is separated from the next by a resistor. At the bottom is the large capacitor for setting the negative voltage and it's diode & discharge resistor. The top right are connections necessary for dividing down the bias voltage for measurement and the bottom left are ribbon cable connections for applying voltage to the tips.

where V_{bi} is the tip bias, $V_+ \geqslant 0$ and $V_- \leqslant 0$ are the applied biases, i is the tip number, and N is the number of tips.

3.3.1.2.2 Problems Although this system works wonderfully to remotely control the bias such that we don't have to be running up and down the stairs all day long, it does come with it's own set of issues.

During running when the probe was at low radius and $V_- = -225$ V, the positive power supply broke and a short to ground was formed



(a) Top view (b) Front view

Figure 3.8: An image of the circuitry for the new bias power supply. In panel 3.8a: the top left is a fan for cooling the charging resistor hidden beneath a heat sink in black, the bottom left shows the negative capacitor that sets the ground of the rest of the circuit, the top right black cylinders are the individual tip capacitors and white rectangles are high power resistors for the voltage division, bottom right are connections to the front panel. In panel 3.8b: the top left is the fan power, the red and black banana connections are to the power supplies, the rainbow ribbon cable is the connection to send bias to the tip circuit, the bottom right black cable is for measurement of the bias voltage, and the green banana is a connection to the laboratory ground.

internal to the power supply between the negative and ground bananda connection terminals. This was most likely due to some transient that sent the positive power supply more than 300 V from ground which the power supply can not do.

Additionally the 10 W discharge resistor initially used for discharging the negative capacitor burned out because the power load was higher than expected. We replaced it with a large carborundum resistor that can handle 50 W.

3.3.1.2.3 Calibration Calibration of the supply consists of two things:

- The power supplies are remotely controlled through LabView which

Name	Symbol	Value	Unit
Old Bias Circuit			
Positive bias voltage	V_+	$0 \leqslant V_+ \leqslant 250$	V
Negative bias voltage	V_-	$-250 \leqslant V_- \leqslant 0$	V
Tip capacitor		1.5	mF
Divider Resistor		???	Ω
New Bias Circuit			
Positive bias voltage	V_+	$0 \leqslant V_+ \leqslant 250$	V
Negative bias voltage	V_-	$-250 \leqslant V_- \leqslant 0$	V
Tip capacitor		1.6	mF
Divider Resistor		1	kΩ
Negative bias capacitor		24	mF
Charging resistor		10	Ω
Discharging resistor		1	kΩ

Table 3.2: Bias circuit component values

communicates with a CompactRIO analog output module that sends an isolated control voltage to the power supply. This output signal has a voltage offset and is not a perfect conversion between set voltage and output voltage that can be corrected by a multiplicative factor. We can do both these corrections by setting a voltage in LabView and seeing what voltage the power supply is outputting. This is just a problem of fitting a straight line with an offset.

- We also need to calibrate the measurement of the bias voltage. Since the digitizers have an internal offset voltage and there is a resistor divider between bias and measurement we must correct for both of those. This is similarly a case of measuring the output bias voltage and correcting the digitizer measurement to match that value.

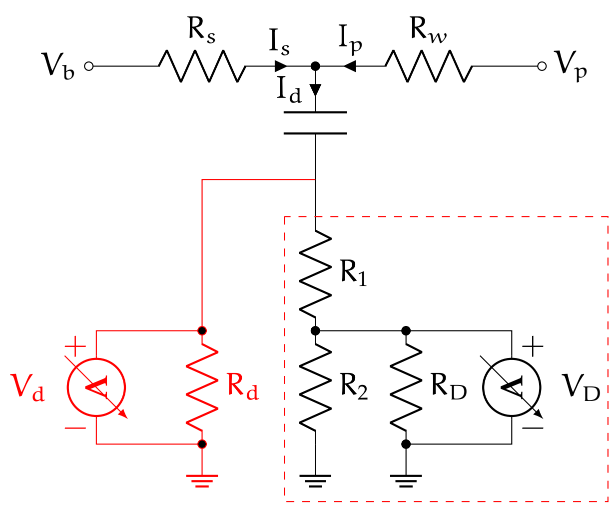


Figure 3.9: Circuit for a single tip on the multitip probe. We can reduce the circuit contained in the red dashed rectangle to the red circuit using the equation for a voltage divider.

3.3.2 Probe Circuit

Now that we can apply a voltage to the tip, we need some way to measure the current to the probe. We do this using a commonly used circuit setup with a few additions for our setup which can be seen in figure 3.9.

3.3.2.1 Principles of operation

We apply the bias at the node labeled V_b. When the probe pulls current from the plasma, I_p, this changes the tip voltage and pulls the voltage closer to the floating potential since there is a voltage drop across the sense resistor, R_s. The capacitor and resistor divider RC times acts to shield the DC component of the probe bias voltage. The voltage difference across the sense resistor is measured by a digitizer which sees V_D but that resistor divider can be simplified to a measured voltage of V_d. We want the probe voltage, V_p, and the probe current, I_p, to fit the plasma parameters.

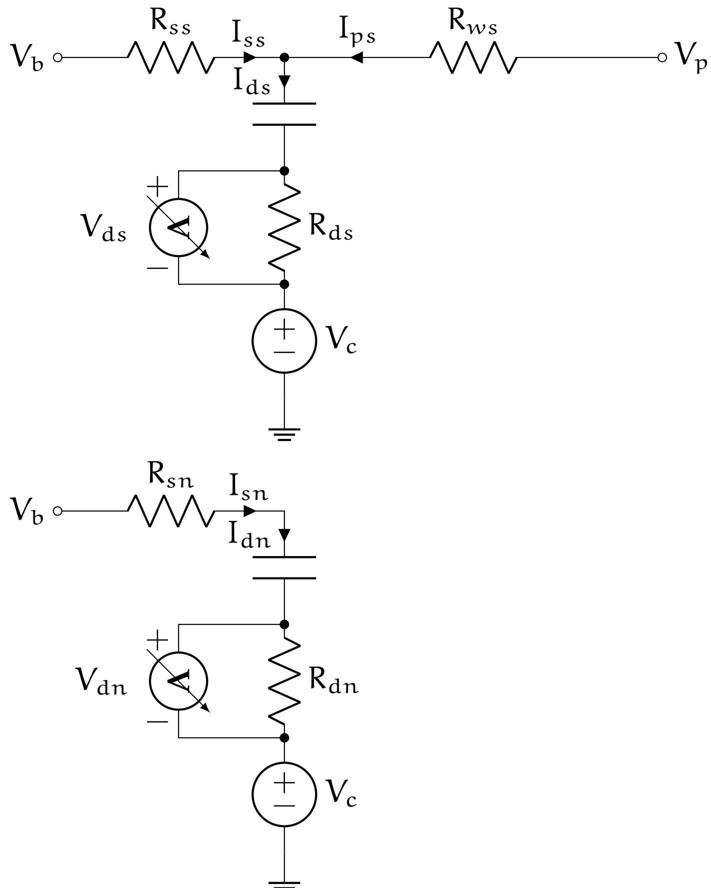


Figure 3.10: Circuit for a single tip on the Langmuir probe alongside the circuit for measuring noise. The V_b inputs are the same for both circuits but since the bias not change during a shot we will ignore any currents flowing between the circuits. The subscripts are organized as (element, circuit) where the elements are b = bias, s = sense resistor, w = wire, p = probe, d = digitizer and the circuits are n = noise, s = signal. In our equations we don't use the raw voltages from the digitizers (V_{ds} and V_{dn}) and instead use the currents (I_{ds} and I_{dn}) which we can find using $V = IR$. The noise is modeled as a changing chassis voltage.

Table 3.3: Probe circuit component values

3.3.2.2 Single-ended measurement

We've used the TREX cells in the past which are single ended digitizers that can sample between 2.5 MHz (32 channels) and 20 MHz (4 channels). These work pretty well for our lower frequency experiments like those with the drive coils. Unfortunately because of the high voltages and large currents, the ground of these digitizers can change a lot during a shot even if they are on isolation transformers to remove their reliance on the lab ground. This circuit can be seen in figure 3.11.

3.3.2.3 Differential measurement

We've developed a new circuit board to make use of our D-TACQ digitizers which measure using a differential input. Instead of measuring between the digitizer ground and the resistor divider output, we instead duplicate all the circuit elements on the board meaning each tip has two sense resistors, two capacitors, and two resistor dividers. One of the circuits is still connected to the probe tip through it's wire but the other is only connected to the bias voltage input. Thus we can measure the difference between these two signals.

This vastly improves the signal quality and helps to remove much of the noise we previously saw when the capacitor banks were triggered.

3.3.2.4 Circuit Solution

During a shot, noise is generated around the lab because of the high currents used to create the reversed magnetic field for reconnection. To remove this noise we use another circuit that has the same components but is not connected to a plasma facing tip which can be seen in figure 3.10. We'll call this other circuit the **noise circuit** and the circuit that is

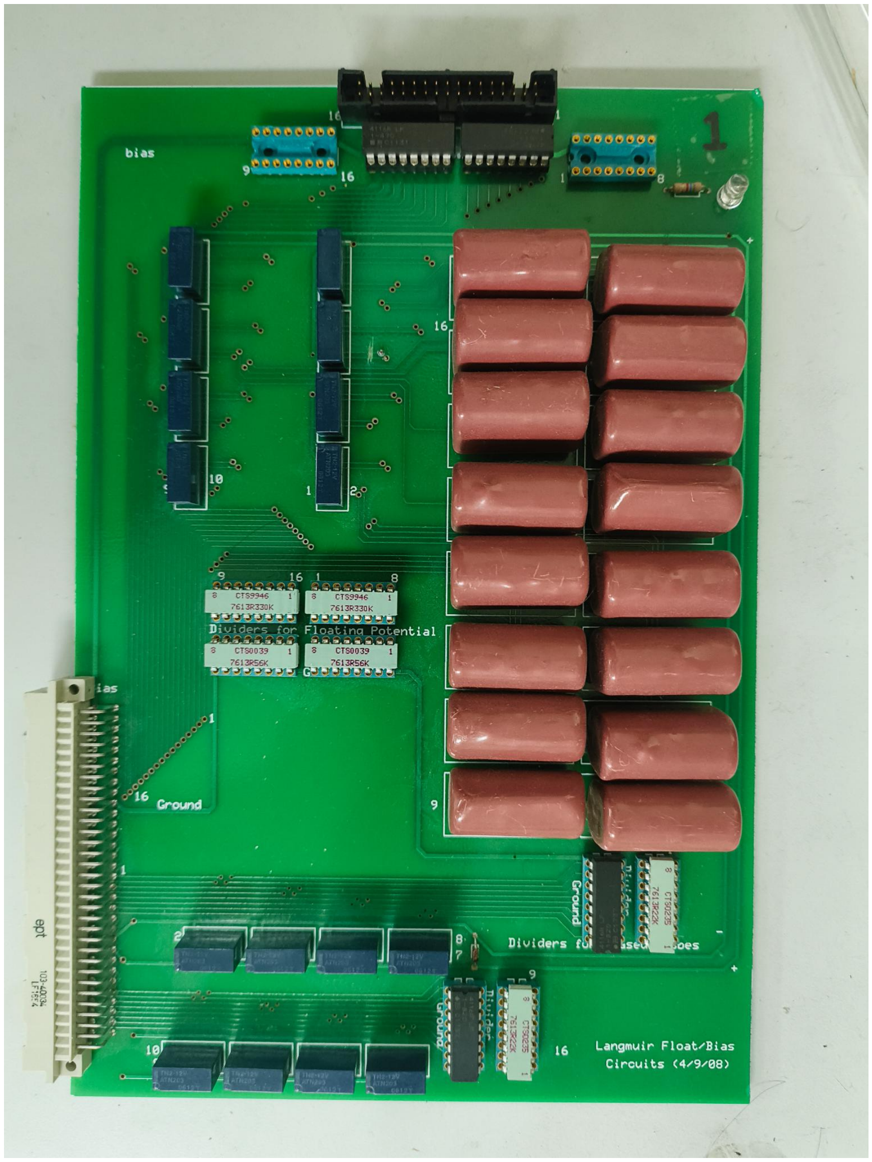


Figure 3.11: Image of circuit board used for single-ended measurements. The top ribbon connector is where the bias comes in. Just below that connector are multiple sense resistors. Often those are replaced with hand soldered higher power resistor networks. The pink rectangles on the right are the capacitors as can be seen in figure 3.9. The resistor networks in black and light blue in the bottom right are a resistor divider prior to measurement whose signals leave out the tan connector at bottom left.

Figure 3.12: Image of new circuit board

connected to the tip the **signal circuit**. If there was no noise in the lab we would expect to measure no fluctuations on the noise circuit digitizer.

In order to get the IV-curve needed for extracting out the plasma parameters we need to solve for the probe voltage and current. To do this, we'll take figure 3.10 as our model where we have modeled the noise as a fluctuating ground voltage[1]. This isn't a perfect model as it ignores other sources of noise from the digitizer or bias fluctuations (which it may actually account for) but it works well for our purposes. The capacitor in this circuit has nearly no time dependence over the shot so we treat it as having a constant voltage drop of V_b.

3.3.2.4.1 How do we solve this Using Kirkoff's laws we can construct a system of linear equations. Kirkoff's current law gives us two equations for the nodes in the signal and noise circuits and Kirkoff's voltage law gives six equations connecting any end node combination in each circuit. We solve these equations using `SymPy`, a symbolic math library in Python. These give the following:

$$\begin{aligned} I_{ps} =& \frac{1}{R_{ss}}[(R_{ds} + R_{ss})I_{ds} - (R_{dn} + R_{sn})I_{dn}] \\ V_{ps} =& \frac{1}{R_{ss}}[(R_{ds}R_{ss} + R_{ds}R_{ws} + R_{ss}R_{ws})I_{ds} \\ & - (R_{dn}R_{ss} + R_{dn}R_{ws} + R_{sn}R_{ss} + R_{sn}R_{ws})I_{dn}] + V_b \end{aligned}$$

Since the digitizers are measuring the voltage output of the resistor divider we have replaced the measured voltages, V_D, with the currents, I_d,

[1]We've also tried modeling the noise using a fluctuating current that appears on both the probe signal and where a probe would attach in the noise circuit. This did not work as well as the ground voltage fluctation method though.

which is the total current flowing through the resistor divider. Since we know the measured voltage and the resistances of the divider and digitizer we can find the current using Fig. 3.9:

$$
\begin{aligned}
R_d &= R_1 + (R_2^{-1} + R_D^{-1})^{-1}, \\
V_D &= \frac{(R_2^{-1} + R_D^{-1})^{-1}}{R_d} V_d, \\
I_d &= \frac{V_d}{R_d}
\end{aligned}
$$

Thus our final equations for the probe voltage and current are

$$
I_d(V_d) = \frac{V_d}{R_d} = \frac{V_D}{\left(R_2^{-1} + R_D^{-1}\right)^{-1}} \tag{3.4}
$$

$$
I_{ps}(I_{ds}, I_{dn}) = \frac{1}{R_{ss}}[(R_{ds} + R_{ss})I_{ds} - (R_{dn} + R_{sn})I_{dn}] \tag{3.5}
$$

$$
\begin{aligned}
V_{ps}(I_{ds}, I_{dn}, V_b) = &\frac{1}{R_{ss}}[(R_{ds}R_{ss} + R_{ds}R_{ws} + R_{ss}R_{ws})I_{ds} \\
&- (R_{dn}R_{ss} + R_{dn}R_{ws} + R_{sn}R_{ss} + R_{sn}R_{ws})I_{dn}] + V_b
\end{aligned} \tag{3.6}
$$

3.3.2.4.2 Assumptions Here we have assumed that all the noise is coming from a fluctuating ground which is an alright assumption as we do expect significant ground fluctuations that could easily change the measurement. We see less of these fluctuations during shots with no plasma which may mean part of this noise is coming from current drawn by the probe. It remains to be seen if there are better ways to model the common-mode noise or if we are accidentally removing bias voltage fluctuations that may be necessary for better IV-characteristic reconstructions.

3.3.2.5 Calibration

Try as we might, we can't perfectly manufacture these probes such that the areas of each tip are the same and the various resistances all match. The best way to calibrate this probe is using the plasma itself.

3.3.2.5.1 Calibrating in Ion Saturation Regime When the current to the probe is dominated by the ions, the IV-characteristic is very flat. By setting all tips to the same very negative bias voltage we measure different currents to each tip. The relative difference between measurements is nearly the same as the relative area difference between those tips.

3.3.2.5.2 Calibrating on the exponential In the ion attracting regime but near the plasma potential, the shape of the IV-characteristic is approximately exponential. Since our initial plasma is very cold (~ 4 eV), the rise is very swift. Thus setting the tip potential to be near the plasma potential during the equilibrium plasma before the drive coils gives us the ability to calculate the wire resistance such that all the points on the IV-characteristic are at the same voltage.

3.3.2.5.3 Bayesian calibration We can actually do both of these calibrations using Bayesian statistics where we solve for the probability of the area (or resistance) based on the current to the probe. By finding the probability of the measured current for some area assuming some amount of noise and then applying a prior that forces the area to be in some resonable range let's us find the most probable area. The same can be done for the resistance.

3.3.2.5.4 Calibration using pressure wave Although we can calibrate the relative areas between probes we need some way to get the actual plasma parameters (not just the relative ones)! We do this using data from

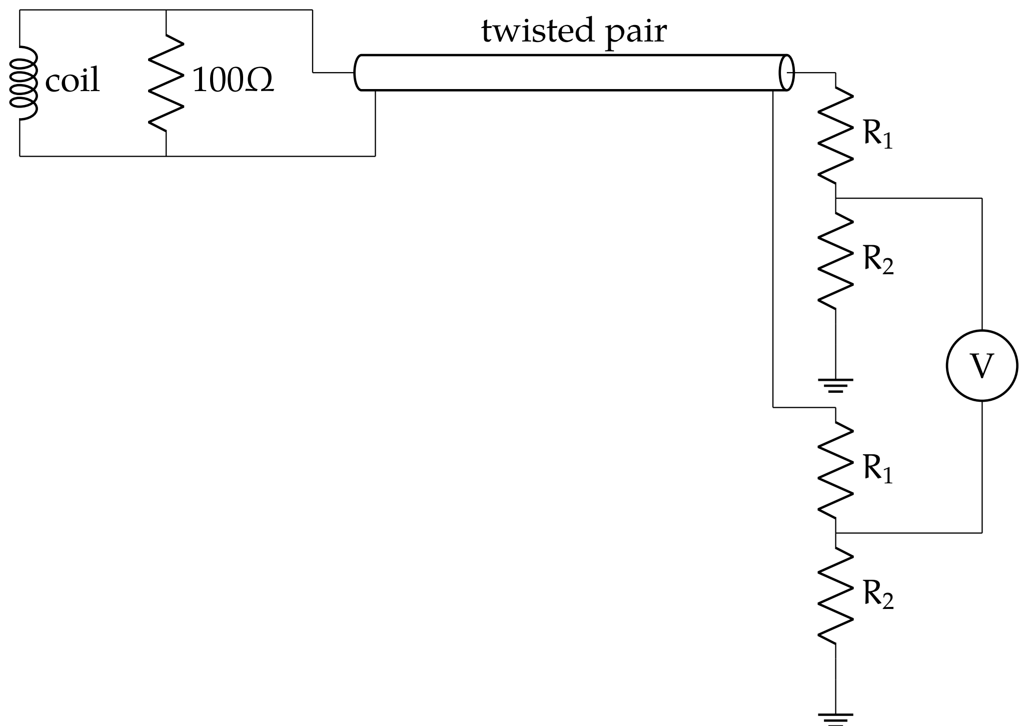


Figure 3.13: The Bdot coil circuit. The top left inductor and resistor and near the probe tip. The signals are extracted using twisted pair and then divided down before a differential measurement.

the previous chapter. Since that method works well to extract the initial density profile we can change the tip areas to best match those results.

3.3.3 Bdot coils

The Bdot coils, while basic in theory, need some help to actually give physical results.

3.3.3.1 In-shaft circuit

Inside the stainless shaft connected to the probe, there is another circuit containing some resistors and various through holes. Since each coil wrap is really two coils (as they are wrapped using twisted pair) this is where those two coil signals are separated. The resistor is in place to improve the system response to high frequency.

3.3.3.2 Circuit model

We model the circuit using figure 3.13. The small resistor near the coil does not come into play for analyzing the signal. As of this writing we assume a linear response between fluctuating magnetic fields and measured voltage but a better solution would be measurement of the transfer function.

3.3.3.3 Calibration

Although we could use a handheld coil for calibrating the areas and directions of this coil, it is far easier and more accurate to use the drive coils as a calibration source. Without plasma, we can model the expected magnetic field assuming they induce current in the BRB walls such that no field can escape (this is true since the BRB shell is made of aluminum and the induced currents have continuous paths). Using these modeled fields we can match the measurement to the expected field and by rotating the probe and moving the probe figure out the vector areas.

3.3.3.4 Signal Interpretation

Once we have the vector area of each coil, $\mathbf{A}_i$, we can now extract the Bdot signal. According to Faraday's law,

$$\mathcal{E}_i = \mathbf{A}_i \cdot \dot{\mathbf{B}}. \tag{3.7}$$

Now we measure a signal, S_i, where

$$S_i = \frac{R_2}{R_1 + R_2} \mathcal{E}_i. \tag{3.8}$$

Assuming the coils make a basis for $\mathbb{R}^3$, we can turn this into an inverse problem:

$$\begin{bmatrix} S_1 \\ S_2 \\ \vdots \\ S_6 \end{bmatrix} = \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_6 \end{bmatrix} \cdot \dot{\mathbf{B}} \tag{3.9}$$

$$\mathbf{S} = \mathbf{A} \cdot \dot{\mathbf{B}} \tag{3.10}$$

Now $\dot{\mathbf{B}}$ is overdefined since we have six measurements and three field components so using linear least squares[2] we can solve for $\dot{\mathbf{B}}$ and also the standard deviation of $\dot{\mathbf{B}}$. This is useful in understanding where we might expect error in our signals and whether we need to smooth signals to get rid of noise caused by the firing of the capacitor banks (which is sometimes needed).

3.4 Plasma Theory

The depth of theory needed to correctly understand Langmuir probes is near endless with a menagerie of exciting physical effects one could account for. I'm prone to dive a bit too deep into some of these various theories but after much literature review and some misteps we've come to a theory that seems to work relatively well for our results. Here I'll discuss the prior method using a very simple model and another method that gives much better results.

[2]We've also tried using generalized least squares (since coils made of the same twisted pair may have more correlated noise) but we did not see a significant improvement.

3.4.1 Simple Plasma Theory (Hutchinson)

Hutchinson (2002b) develops a simple model for the current to a Langmuir probe that we have used for a long time. It assumes that the ion current is constant and the electron current follows an exponential and only looks at the electron repelling region.

3.4.1.1 Derivation

We won't reproduce the derivation here but will instead describe the physical interpretation.

The volume surrounding the probe is split into two regions: **the plasma** where the electric field is very small, quasineutrality holds, and the potential is near the plasma potential vs. **the sheath** where the ion electron charge densities are different and the electric field is largs such that the potential quickly jumps to the probe potential.

The ion current is just modeled by how many ions randomly cross the plasma-sheath interface (the interface will be denoted by subscript s). If the ions are cold (i.e. negligible initial kinetic energy) then they will only get to the probe via their random motion. The speed of these ions is set by the voltage at which the interface appears which is where

$$V_s = -\frac{T_e}{2}. \tag{3.11}$$

V_s is the potential at the interface relative to the plasma potential and T_e is in eV. Assuming the electrons are Boltzmann and quasineutrality holds at the interface then

$$n_{is} = n_{es} = n_\infty \exp\left(\frac{V_s}{T_e}\right). \tag{3.12}$$

Now we can calculate the ion current as

$$
\begin{aligned}
I_i &= A_s v_{is} n_{is} && (3.13)\\
&= A_s \sqrt{-\frac{2eV_s}{m_i}} n_\infty \exp\left(\frac{V_s}{T_e}\right) && (3.14)\\
&= \exp\left(-\frac{1}{2}\right) A_s n_\infty \sqrt{\frac{eT_e}{m_i}}. && (3.15)
\end{aligned}
$$

We often assume that the sheath area is the same as the probe area since the Debye length is so short.

For the electron current, we are assuming they are Boltzmann and they will hit the probe due to random motion. Since they are repelled by the probe the density will be smaller than in the bulk plasma. We can write the electron current as

$$
I_e = A_p \frac{1}{4} \sqrt{\frac{2eT_e}{m_e}} n_\infty \exp\left(\frac{e(V_{\text{probe}} - V_{\text{plasma}})}{T_e}\right) \tag{3.16}
$$

where the square root is the average electron velocity and the exponential is the factor the electron density is reduced.

The total current to the probe for $V_{\text{probe}} < V_{\text{plasma}}$ is thus

$$
I = n_\infty e A_p \sqrt{\frac{eT_e}{m_i}} \left[\frac{1}{2}\sqrt{\frac{2m_i}{\pi m_e}} \exp\left(\frac{V_{\text{probe}} - V_{\text{plasma}}}{T_e}\right) - \exp\left(-\frac{1}{2}\right)\right]. \tag{3.17}
$$

3.4.1.2 Pros & Cons

This method has the benefit of being exceedingly simple to implement and often widely applicable. As long as the tips stay below the plasma potential this usually nicely matches our measurements. Unfortunately there's the catch, during a shot the plasma potential changes quickly by up to 100 or more volts. One option is to select only probe tips that look like they are below the plasma potential but this is both time consuming and prone to bias. Thus we look for a new method.

3.4.2 BRL

Laframboise (1966) created a model for the current to a cylindrical probe in a plasma with maxwellian ions and electrons that works for probe voltages above and below the plasma potential. Unfortunately, to my knowledge nobody has redone these calculations nor made a public repository with the code. To sidestep this issue we've used two simplifying approximations.

For probe potential greater than plasma potential, i.e. electrons attracted, we follow along with appendix F of Laframboise (1966). We assume the ions have zero temperature. Although we have no way currently to calculate the ion temperature it is an assumption we commonly make for plasmas from our plasma guns. Additionaly we assume we assume that the probe radius is far larger than the debye length. This approximation is valid for the initial plasma but can change throughout the shot. We also assume that all particles that touch the probe are absorbed by it.

Since the scale over which the potential changes is proportional to the debye length, we assume we have a planar probe. Thus Poisson's equation is given by

$$\frac{d^2\varphi}{dx^2} = -\frac{\rho}{\varepsilon} \tag{3.18}$$

Since the ions have zero temperature, they can not enter the $\varphi(x) > \varphi_0$ region and thus for the plasma charge density, $\rho(x)$, they can be ignored. Without loss of generality we'll set the plasma potential, φ_0, equal to 0. Since this region of electrons only plasma physically can't extend to infinity let's define $x = 0$ to be this boundary and let x increase towards the probe.

As this is a one-dimensional system then we have three conserved quantities. v_y and v_z which denote the transverse velocity will not be changed by the potential and the kinetic energy in the x direction along with the electrostatic potential, $E_x = mv_x^2/2 + e\varphi(x)$. e is the elementary charge. As all electrons will be absorbed by the probe, we will only have flow into the $\varphi(x) > 0$ region. Thus our distribution function looks like

$$f(x, E_x, E_t) = \begin{cases} n_0 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{E_x(x)}{kT}\right) \exp\left(-\frac{E_t}{kT}\right) & v_x > 0 \\ 0 & v_x < 0 \end{cases} \tag{3.19}$$

We can now write the density as

$$\begin{aligned} n(x) =& \int_0^\infty dv_x \int_{-\infty}^\infty dv_y \int_{-\infty}^\infty dv_z f(x, E_x, E_t) \\ =& \int_0^\infty \frac{dv_x}{dE_x} dE_x \int_{-\infty}^\infty dv_y \int_{-\infty}^\infty dv_z f(x, E_x, E_t) \\ =& n_0 \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^\infty \frac{\exp\left(-\frac{E_x(x)}{kT}\right)}{\sqrt{2m(E_x - e\varphi(x))}} dE_x \\ & \int_{-\infty}^\infty \exp\left(-\frac{mv_y^2}{2kT}\right) dv_y \int_{-\infty}^\infty \exp\left(-\frac{mv_z^2}{2kT}\right) dv_z \\ =& n_0 \sqrt{\frac{m}{2\pi kT}} \int_0^\infty \frac{\exp\left(-\frac{E_x(x)}{kT}\right)}{\sqrt{2m(E_x - e\varphi(x))}} dE_x \end{aligned}$$

Let's define $\eta = n(x)/n_0$ where n_0 is the number density at infinity. Let's also define $\chi(x) = e\varphi(x)/kT$ and $\beta = E_x/kT$ as the normalized potential and energy respectively. Thus we can write the normalized density as

$$\begin{aligned} \eta(x) &= \frac{1}{2\sqrt{\pi}} \int_0^\infty \frac{\exp(-\beta)}{\sqrt{\beta - \chi(x)}} d\beta \\ &= \frac{e^{-\chi(x)}}{2} (1 - \mathrm{erf}(\sqrt{-\chi(x)})) \end{aligned} \tag{3.20}$$

Using our now normalized quantities, let's plug them into Poisson's equation 3.18 to find

$$\begin{aligned}
\frac{\varepsilon kT}{e}\frac{d^2\chi}{dx^2} &= e\eta(x) \\
\lambda_D^2\frac{d^2\chi}{dx^2} &= \eta(\chi(x)) \\
\frac{d^2\chi}{ds^2} &= \eta(\chi(s))
\end{aligned} \tag{3.21}$$

where $s = x/\lambda_D$ and $\lambda_D = \sqrt{\varepsilon kT/n_0 e}$ is the Debye length in the undisturbed plasma.

Since the potential is zero at the sheath boundary and the derivative is smooth, $\chi(0) = 0$ and $d_s\chi(0) = 0$. For all $s > 0$, the density of electrons should be non-zero, i.e. we should never achieve a full vacuum except at infinity. Thus,

$$\frac{d^2\chi}{ds^2} > 0,\ \frac{d\chi}{ds} = 0 \Rightarrow \frac{d\chi}{ds} > 0 \Rightarrow \left(\frac{d\chi}{ds}\right)^{-1} = \frac{ds}{d\chi}$$

This means

$$
\begin{aligned}
\frac{d^2\chi}{ds^2} &= \frac{d}{ds}\left(\frac{ds}{d\chi}\right)^{-1} \\
&= \frac{d\chi}{ds}\frac{d}{d\chi}\left(\frac{ds}{d\chi}\right)^{-1} \\
\eta(\chi) &= -\left(\frac{ds}{d\chi}\right)^{-3}\frac{d^2s}{d\chi^2} \\
\Rightarrow \int_0^{\chi}\eta(\chi')d\chi' &= \frac{1}{2}\left(\frac{ds}{d\chi}\right)^{-2} \\
\Rightarrow \frac{ds}{d\chi} &= \frac{1}{2\int_0^{\chi}\eta(\chi')d\chi'} \\
\Rightarrow s(\chi) &= \int_0^{\chi}\frac{d\chi'}{2\int_0^{\chi'}\eta(\chi'')d\chi''}
\end{aligned}
$$

This integral can be evaluated numerically. Thus we can pass a probe potential, χ, and after integrating get the distance from the probe to the start of the sheath.

Since all particles that enter the sheath are collected, from equation 3.19, then we need only find how the surface area of the sheath changes for different potentials. Letting I_0 be the current collected from the electrons when the probe is at the plasma potential then the current to the probe is

$$I(\chi) = I_0\left(1 + \frac{\lambda_D}{R_p}s(\chi)\right)^2 \tag{3.22}$$

Now that we have a model for the current to the probe for the electron attracted region, we move on to modeling the current to the probe for the ion attracted region. There has been much more work put into this region from Steinbrüchel et al. Steinbrüchel (1990) Karamcheti and Steinbrüchel (1999), Mausbach Mausbach (1997), and Chen Chen (1965). Steinbrüchel fitted a model of the form

$$I = I_{i0} a(-\chi)^b + I_{e0} \exp(\chi) \quad \chi < 0$$

where a and b were fit to the data given by Laframboise. Mausbach used

$$I = \begin{cases} I_{i0} a_i (b_i - \chi)^{c_i} + I_{e0} \exp(\chi) & \chi < 0 \\ I_{e0} a_e (b_e + \chi)^{c_e} & \chi \geqslant 0 \end{cases}$$

where $a_{i,e}$, $b_{i,e}$, and $c_{i,e}$ are fitted parameters for the ions and electrons.

Unfortunately neither of these correctly predict the ion current to the probe for large χ so we instead use the parametrization given by Chen Chen (2001) which better predicts ion current for $\chi < 0$. The total current to the probe is given by

$$\begin{aligned} I &= I_{i0} i(\chi) + I_{e0} \exp(\chi) \\ \frac{1}{i^4} &= \frac{1}{(A\chi^B)^4} + \frac{1}{(C\chi^D)^4} \end{aligned}$$

where A, B, C, and D are functions of $\xi \equiv R_p/\lambda_D$. We will reproduce the work from Chen here for the sake of completeness

ABCD are defined in two different regimes. For large ξ ($\xi > 3$),

$$\begin{aligned} A, B, D(\xi) &= a + b(\xi - c)^d \exp[-f(\xi - c)^g] \\ C &= a + b \exp[-c \ln(\xi - d)] + f(1 - g \ln \xi). \end{aligned}$$

This is accurate to within 3%.

		a	b	c	d	f	g
$\xi > 3$	A	1.142	19.027	3.000	1.433	4.164	0.252
	B	0.530	0.970	3.000	1.110	2.120	0.350
	C	0.000	1.000	3.000	1.950	1.270	0.035
	D	0.000	2.650	2.960	0.376	1.940	0.234
$\xi < 3$	A	1.12	.00034	6.87	0.145	110	—
	B	0.50	0.008	1.50	0.180	0.80	—
	C	1.07	0.95	1.01	—	—	—
	D	0.05	1.54	0.30	1.135	0.370	—

Table 3.4: Values for $abcdfg$ in the Chen parametrization of the ion current to a probe.

Although not as accurate but useful for all ξ, we have

$$
\begin{aligned}
A &= a + \frac{1}{\frac{1}{b\xi^c} - \frac{1}{d\ln(\xi/f)}} \\
B, D &= a + b\xi^c \exp\left(-d\xi^f\right) \\
C &= a + b\xi^{-c}
\end{aligned}
$$

The values for a, b, c, d, f, and g are given in table 3.4.

3.4.3 Effect of electric fields

Electric fields during reconnection can change the average plasma potential all of the tips see which is extracted by fitting the plasma theory to the tip measurements. The effect of these electric fields is non-local because the measured plasma potential is integrated along the path of the probe shaft.

This isn't all though; electric fields can also have a local effect directly measurable by the difference in measurement between tips on one side of the probe vs. the other (see figure 3.3).

3.4.3.1 Where does the electric field come from

There are two types of electric field to worry about:

During reconnection we expect the magnetic field to be quickly changing along the axis of the machine as the z directed field reconnects. Since we are dealing with a plasma we expect small static electric fields because of quasineutrality. Also, since the system is toroidally periodic there can be no net static field in the ϕ direction.

3.4.3.1.1 Inductive Fields These are generated by the change in magnetic field as the experiment progresses. Let's discuss what each direction magnetic field change can create:

- z: This is the field that changes most during the experiment because it is reconnecting. Because of toroidal symmetry, the most general closed loop we can construct are annulus arcs. Because of that same symmetry the changing z field will create a negligible r directed electric field but can create toroidal electric fields (in fact the drive coils generate an immense toroidal electric field to drive reconnection).

- r: This field can create either a z or ϕ directed electric field. The ϕ electric field can be generated but once again, because of symmetry, the closed loop on the boundary of a cylinder arc causes E_z to be minimal. One other detriment to E_z is the small magnitudes of B_r which can form in the outflow region of reconnection but are not large.

- ϕ: The source of ϕ field can come from two places: a guide field that is frozen into the plasma (see 1.3.2.3) or the hall fields when the ions and electrons decouple (see 1.2.4). This allows for the formation of both r and z electric fields but because of the often small magnitude of B_ϕ we don't expect too large of fields.

3.4.3.1.2 Static Fields Since our plasmas are often quasineutral we don't expect too huge of static electric fields. Additionally, because of toroidal symmetry there can be no net static electric field in the ϕ direction because if there was a net electric field this would force the electrostatic potential to be multivalued which is physically impossible. We do have a couple places where it can appear though.

As discussed in 1.2.5.3, a trapping potential forms that changes the shape of the flux tube as it moves towards the x-point during reconnection. This potential is on the order of a few T_e Egedal et al. (2009) which gives an electric field in our device on the order of ??? V/m. This trapping potential forms to create electric fields in the rz-plane.

During the pressure wave, the low inertia of the electrons let's them speed up far faster than the ions. This acceleration occurs as the wavefront passes the plasma. This generates an ambipolar electric field in the $+r$ direction that slows the electrons and accelerates ions to enforce quasineutrality.

3.4.3.1.2.1 What is the expected static magnitude

3.4.3.2 What does it do to the tip measurements

Let's first understand what we expect to happen and then look at some measurements.

3.4.3.2.1 Theoretical understanding Figure 3.14 shows what we expect the electric field to do to the tip biases. Any electric field, whether it be inductive or static, can cause the tip voltages to be different than what we expect from the circuit model.

3.4.3.2.2 Physical reality We can see a measurement from a shot where the probe tips were pointed in the ϕ direction (and thus seeing the effect

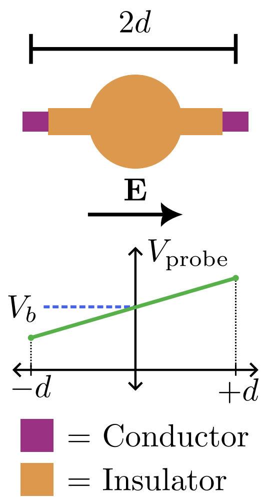


Figure 3.14: The expected effect of an electric field on the probe measurements. The tips are separated across the probe body by a distance 2d. There is an electric field of magnitude E pointed along the direction of the tips. If both tips are set to a bias V_b, the electric field will change the left tip to be at $V_b - Ed$ and the right tip will be at $V_b + Ed$.

of E_ϕ) in figure 3.15. It can be seen that tips on one side of the probe are shifted compared to tips on the other side (even and odd numbering can be seen in 3.3).

3.4.4 Minimum measurable density

Since we have many tips next to each other, we may need to worry about them interfering with each other and changing the IV-characteristic. This interference is directly related to the Debye length.

3.4.4.1 What's the Debye Length

The Debye length is the distance at which electric fields are shielded in a plasma and goes as

$$\lambda_D = \sqrt{\frac{\epsilon_0 T_e}{ne}}. \tag{3.23}$$

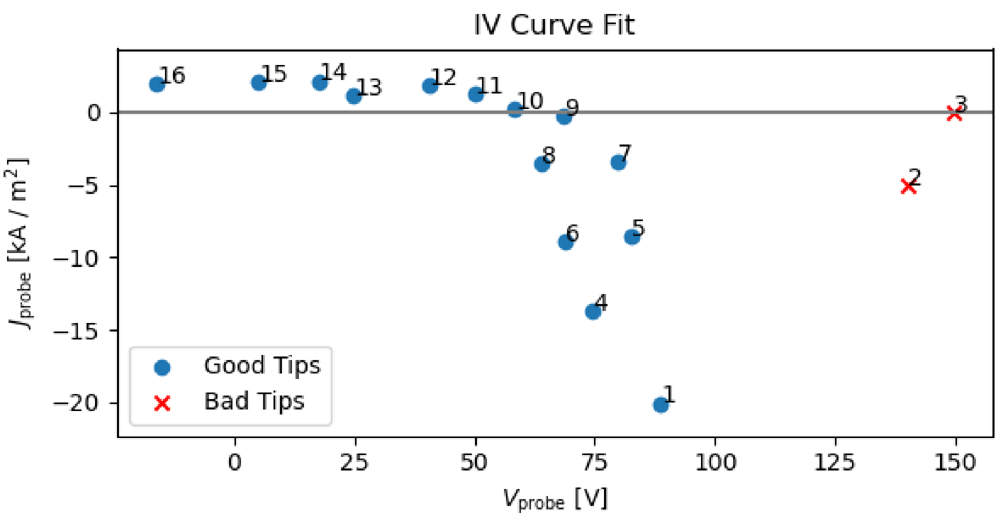


Figure 3.15: Example JV-characteristic showing the splitting of tips that are on one side of the probe vs. the other (even vs. odd). It can be seen that the even numbered tips are moved to a negative bias relative to the odd numbered tips. The actual plasma potential is at the midpoint of the shifts left and right.

Electric fields drop exponentially like $E = A\exp(x/\lambda_D)$ because the electrons and ions are somewhat free to move and will quickly remove any static electric field. Inductive fields on the other hand stick around and instead cause the plasma and it's frozen in magnetic field to move.

3.4.4.2 Sheath scale

As discussed in 3.4.1.1, one way to model a Langmuir probe is to use a sheath-plasma interface where quasineutrality holds in the plasma but electrons and ions separate inside the sheath. Although not all Langmuir theories have it (e.g. BRL), this interface is usually present to some approximation. The distance at which this sheath appears is often on the order of $10\ \lambda_D$.

3.4.4.3 Tip-tip sheath overlap

If we have sheaths of neighboring tips that overlap (either because of high temperature of low density from equation 3.23) this could change the IV-characteristic that we measure. Often once we have high temperatures the density is high enough such that the Debye length is far smaller than the tip separation (see table 3.1) so we won't worry about temperature effects. Instead, early in the shot when the density can be low the sheaths can start to overlap. Assuming the sheath length is $20 \times \lambda_D$, this gives a minimum measurable density of

$$x = 20\lambda_D \tag{3.24}$$

$$= 20\sqrt{\frac{\epsilon_0 T_e}{n_{min} e}} \tag{3.25}$$

$$\Rightarrow n_{min} = \left(\frac{20}{x}\right)^2 \frac{\epsilon_0 T_e}{e} \tag{3.26}$$

$$\approx 2e - 16 \text{ m}^{-3} \tag{3.27}$$

where we've set $T_e = 4$ eV which is often what is seen in the initial equilibrium and x is the distance between tips.

3.5 Curve Fitting

Once we have measurements of the probe voltage and current we can extract the plasma parameters. This is done using some sort of fitting routine.

A commonly used fitting routine is by subtracting off the ion current in the ion attracted regime and then fitting the exponential in log space using a straight line. This works quite well in swept Langmuir probe applications where you have many points on the IV-characteristic and there is not much noise.

Our system doesn't fit those criteria at all so we need to try some other options.

3.5.1 Least squares Fitting

Least squares has often been used in our lab because of it's simplicity. This fitting routine operates by minimizing the squared error in J between the function to fit, $J(V)$, and the samples, (V_i, J_i).

3.5.1.1 Benefits

Least squares is very quick to solve and thus analyzing many data points is no issue. Additionally, it is extremely easy to implement as just abaout any programming language offers this as an initial form of model fitting.

3.5.1.2 Issues

Unfortunately it's ease also has it's issues. Since it is only minimizing error in the J direction it does not account for errors in measurement that also cause errors in V. This is especially problematic when using Hutchinson's model and the probe is collecting in the electron saturation region. Since Hutchinson's model does not account for this component it artificially raises the electron temperature and this is especially visible when only minimizing the error in J. Least squares also often uses a simple method for finding a 'best fit'. This means that if the fit starts with a bad initial guess it may end at a suboptimal fit.

3.5.2 ODR

To correctly model the errors in our circuit we use Orthogonal Distance Regression (implemented by Boggs et al. (2005)).

3.5.2.1 What does error look like in our signals

We use three different measurements for calculating the probe current and voltage and multiple parameters that we set for the circuit. Although these parameters need calibration we don't account for errors in them and instead set them to some static value for all shots.

Our measured parameters are V_{ds}, V_{dn}, and V_b which can be seen in equations 3.4-3.6. Errors in our V_b measurement will only cause errors in V_p whereas V_{ds} and V_{dn} will cause shifts mostly along a line that goes like

$$\Delta_I = \frac{R_d + R_s}{R_d R_s + R_d R_w + R_s R_w} \Delta_V.$$

3.5.2.2 How does ODR take that into account

Instead of passing the final values to fit, (V_i, J_i), we instead pass the raw measurements, $(V_{Dsi}, V_{Dni}, V_{bi})$. Then we are trying to fit the model defined by

$$\begin{aligned} 0 &\approx f(n, T_e, V_p; \mathbf{V}_{Ds}, \mathbf{V}_{Dn}, \mathbf{V}_b) \\ &= J_p^{\text{theory}}(V_p^{\text{circuit}}(\mathbf{V}_{Ds}, \mathbf{V}_{Dn}, \mathbf{V}_b), n, T_e, V_p) - J_p^{\text{circuit}}(\mathbf{V}_{Ds}, \mathbf{V}_{Dn}). \end{aligned}$$

Here the superscript defines whether we are using the plasma model (which depends on the plasma parameters to fit) or the circuit model (which depends on the circuit parameters and measurements). We've converted everything to current density as that removes ambiguity with dividing the probe current by the relative areas between tips.

Now as ODR tries to fit this model, it'll allow error in all of our measured values which correctly reconstructs the errors we might expect from our measurements.

3.5.2.3 How to apply parameter bounds

More recent iterations of ODR have methods of applying bounds Zwolak et al. (2007). Unfortunately because of dependency issues between MDSPlus (our experiments data storage system) and `odrpack` we use a different method for applying bounds.

3.5.2.3.1 Why apply bounds It's not clear that we necessarily need bounds for the parameters we are trying to fit. Why not let the plasma parameters wander to wherever they best match the data? Well, if we did that we would get a lot of spurious solutions that wouldn't be continuous in time and also potentially unphysical (what does a negative density mean? a negative temperature? what if the fit temperature is 10^{10} eV?).

3.5.2.3.2 Our method We know that physically the density and temperature should both be greater than zero. An easy way to do this is to instead fit the logarithm of those parameters. Then no matter what the final fit value is, when converted to a physical number it will be positive.

Sometimes we want to set both upper and lower bounds instead of just lower bounds. To do this we fit a parameter x for the plasma parameter y where

$$y = y_{\text{min}} + (y_{\text{max}} - y_{\text{min}}) \left(\frac{1}{2} + \frac{1}{\pi} \arctan(x) \right).$$

Thus no matter the final fit value of x, the plasma parameter stays within the set bounds.

3.5.3 Monte-carlo Fitting

The dream of many (or at least me) is an algorithm that always finds the global minimum to a given function. Since not all dreams come real we use a simple method that will do instead.

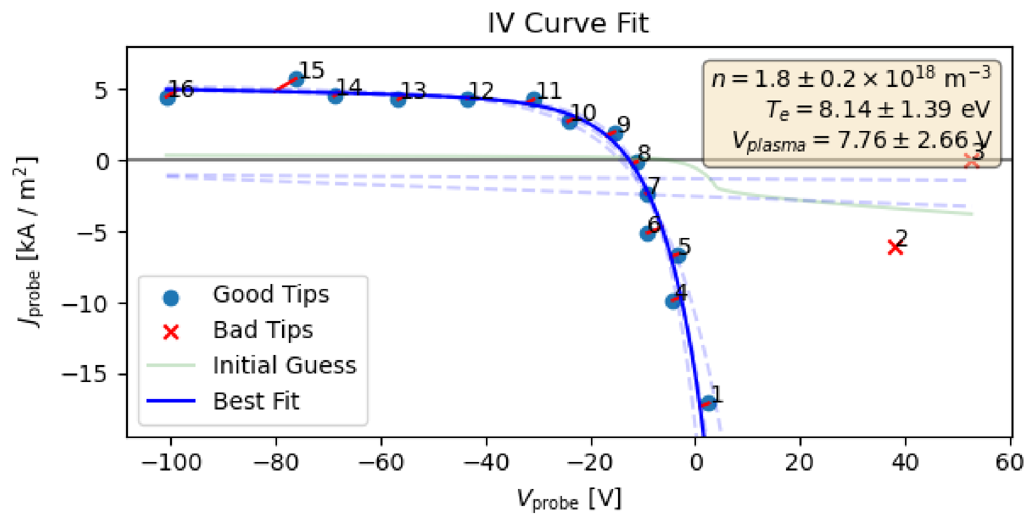


Figure 3.16: Example fit for using Chen's IV-characteristic and Orthogonal Distance Regression.

3.5.3.1 Global vs. local minima

What do I mean by a global minimum though and why does it matter. As can be seen in figure 3.16, the dashed blue lines are 'local' optimums to the fitting function. That is the fitting function thinks it has found the correct solution and stops fitting. An example of this is Newton's methon for a function with many bumps. A starting point may or may not lead to the global minimum and could instead be stuck elsewhere. As can be seen from the figure some of those solutions are really bad (like the ones that are nearly horizontal lines).

3.5.3.2 A simple solution

A bad starting point can lead to a bad solution so let's do something simple: let's take lots of starting points! Although we may get some attrocious solutions, some of them will match much better with the data and we can choose the best option.

3.5.3.3 How do we sample initial conditions

We start the procedure with a best initial guess. This is often whatever the previous fitting result is because the plasma should be somewhat similar to the plasma 0.1 microseconds ago. Now we choose some amount that the plasma may have changed by (i.e. the density could be an order of magnitude smaller or an order of magnitude higher). Now we have starting parameter bounds on n, T_e, and V_{plasma}. We choose some number of starting points N. Now we use latin hypercube sampling which tries to well sample the space of possibilities without overlapping parameter choices.

3.5.3.4 Choosing the best answer

Now that we have N possible solutions we just look for the solution that minimizes the measurement error the most. Simple as can be.

3.5.3.5 More robust methods for getting global minimums

Another option than our simple multiple starting point method is something like differential evolution which samples many starting points and iteratively finds regions where fits end up better and better. Although we could implement this it would be very slow but should you have infinite computing time, let me know how it goes!

3.5.4 Combining multiple shots

During the course of a shot, the plasma potential can jump by more than 150 V. We want to sample many points near the exponential which we want all the tips to have similar V_{probe}. Thus our solution is to combine multiple shots where the probe tips are shifted by some amount each shot so for one shot we can handle low V_{plasma} and for another we can deal with high V_{plasma}.

Figure 3.17: Example fit using many shots, multiple local parameter fits, and errors

3.5.4.1 Using some time points from each shot

One option for this method is to look at each shot with the same experimental parameters and the probe at some location and select time regions to use. This is nice in that you aren't averaging over multiple shots but can make the plasma parameters jump from one solution to another. Also you are losing potential data that could be useful in getting a more accurate fit.

3.5.4.2 Combining all IV points and fitting that

Another option is to combine multiple shots and fit all the tip measurements from all the tips at once. This has the advantage of getting more accurate fits and a more continuous plasma parameter solution both of which are desirable.

There are two issues with this:

- Lack of reproducibility smears out the curve leading to eronious plasma parameter solutions. This is difficult to deal with in either case as one arbitrarily chooses one shot and the other tries to combine them in a potentially strange manner.
- Our triggering method is quite accurate but has a standard deviation on the order of a couple tenths of microseconds. This requires some timing alignment which we will go into in 4.5.1.1.

3.6 Results

What is the benefit of doing all this work?

Figure 3.18: Plot of isat currents from BRL

Figure 3.19: Radial chord of initial density with Hutchinson vs BRL.

Figure 3.20: Comparison of expected E from Ohm's law and measured E.

3.6.1 Measurement of low densities

3.6.1.1 How does Hutchinson vs BRL treat ion saturation

3.6.1.2 What are the experimental predictions that Hutchinson vs BRL have

3.6.1.3 How does that change the results

3.6.2 Noise resilience

Not sure how best I should show this.

3.6.3 Fits more accurate

- Each fit uses many data points
    - Don't have to arbitrarily cutoff data points

3.6.4 Extracting electric fields

3.6.4.1 Where do we expect electric fields to be

3.6.4.2 Do we see them there

4 PRESSURE ANISOTROPY PROBE

Pressure anisotropy, as discussed in 1.2.5.2, is a difference between the average kinetic energy parallel to the magnetic field vs. perpendicular to the field. During reconnection, our interest lies in understanding the electron anisotropy as simulations show that it can have major impacts on the reconnection dynamics Ohia et al. (2012).

4.1 How can we measure Pressure Anisotropy

We need some way of actually measuring the electron distribution function or at least the parallel and perpendicular temperatures so let's examine a number of possibilities.

4.1.1 Thomson Scattering

Thomson scattering has long been used in the fusion plasma community for measuring electron distribution functions.

4.1.1.1 Basics

Thomson scattering diagnostics use the Thomson scattering effect (a astutely named diagnostic) where a high energy laser scatters off of electrons. By measuring the spectrum of the scattered radiation for a single direction, the distribution function in that direction can be attained. Then, if the measurement direction is moved a new direction in the distribution function can be measured.

4.1.1.2 Pros

Thomson scattering is great in that it can be very localized (just based on the optics of the beam and the collection). Our system is not so reproducable that this high localization is necessary but it is nice nevertheless.

Since Thomson scattering usually operates in a pulsed fashion, the timing and duration of pulses can be highly controlled to allow for very temporally localized measurements as well.

It's also non-perturbative (except for the tiny effect it has on few electrons) since it's just a laser passing through the plasma. This is especially useful in a fusion device where physical probes are impossible. One nice thing about using lasers is there no need to build long probes (except potentially to mount collection optics on).

4.1.1.3 Cons

Unfortunately there are number of problems with Thomson scattering that we would need to face should we attempt to implement it:

- Thomson scattering requires a lot of work with optics and our lab lacks that expertise as of this writing. The startup cost for the laser and all the hardware needed would be large along with the training/learning required to get the system up and working.

- A classic problem with Thomson scattering is getting enough photons to get through the noise. Because of the size of our machine and the relatively low densities, this would only increase the difficulty.

- We want many measurements of the plasma distribution in time to see how it evolves across the layer. Thomson scattering is often usually sampling at 1 kHz and even systems where pulses can be more squeezed together are still very far apart in time relative to our timescales.

- Since our system triggers at slightly different times for each shot and the laser system would most likely only fire once per shot, it would be very difficult to get good localized resolution of measurement.

- The low temperatures of our system (on the order of 10s of eV) necessitate the use of high spectral purity lasers or measurement optics. Although not impossible this raises the cost of setup and reduces the laser options making everything more difficult.

4.1.2 Retarding Field Energy Analyzer

A retarding field energy analyzer (RFEA) uses a set of grids at different potentials to repel all particles up to some kinetic energy. They can be used for either electrons or ions but here we'll discuss electron RFEAs.

4.1.2.1 Basics

RFEAs can come in a variety of geometries but here we'll discuss a planar RFEA with a hole where particles can enter. Particles initially pass through an aperture such that particles mostly from a single direction are measured, then ions are repelled using a grid at a high positive potential, then electrons of a minimum energy are selected by another grid at a negative potential, and finally electrons are collected and the current is measured.

4.1.2.2 Pros

RFEAs are quite good at extraccting the tail of a distribution because the higher the repelling voltage, the more you are isolating out only high energy electrons. The electron distribution is extraced by taking the derivative of the measured current with respect to the applied voltage at a single time. For our configuration we would use a static repelling potential because sweeping the voltage in time would not be feasible in the experiment timescales. RFEAs have been shown to work to very high sampling rates so that would not be an issue for us Hu et al. (2017).

4.1.2.3 Cons

Since RFEAs need multiple grids they are usually somewhat sizable (around a centimeter on all sides). In the case of creating a multidirectional probe, which would vastly accelerate data collection, the size of the probe would become quite large and non-local and perturbative physics may come into play for understanding the data. Since our plasma potential changes so drastically during a shot, a measurement of that potential would be nice to have for RFEA interpretation which could be done with a separate probe but would require a larger probe.

Overall, an RFEA is a promissing opttion especially for some of the other physics we are interested in during reconnection but we settled on a Langmuir probe because of past experience and simplicity.

4.1.3 Langmuir Probes

We've already described the basics of a Langmuir probe in section 3.1 so here we will discuss some of the most relevant parts.

4.1.3.1 Basics

Langmuir probes can come in many configurations, each optimized for measuring a certain plasma parameter. Some of the most common setups are

- A single tip cylindrical probe which is just a long thin wire. This is useful for measuring the overall IV-characteristic from which plasma parameters can be extracted either by fitting or taking the derivative of the probe current with respect to voltage (specifically for getting the electron distribution function) Godyak and Demidov (2011).

- Two plates pointed in opposite directions (like that seen in fig. 3.5) which can measure the ion drift velocity by measuring the ratio of ion current to one plate vs. the other Chung (2012).

- A sheild and a collector setup where the collector is retracted inside the shield such that only a limited number of electrons can hit the probe because their gyroradii force them to instead hit the shield. This can be used to balance the electron and ion currents such the plasma potential can be directly measured but requires high magnetic fields.
- A directional probe where an alumina sheath around a cylindrical probe has a hole perpendicular to the axis of the cylinder. The hole is rotated between shots to collect at different angles relative to the plasma flow and the ion saturation current is measured.

4.1.3.2 Pros

We've already discussed the use of a multi-tip Langmuir probe that has worked well to extract the plasma parameters isotropically (meaning it is collecting plasma from all angles) and thus we have a fair bit of experience with these types of probes. Additionally these probes can be built to almost any size we may want so size considerations are less of an issue. These probes are very cheap to build and can also be moved relatively easily as compared to an optical system.

4.1.3.3 Cons

Utilizing Langmuir probes for directional measurements of the electron distribution has not yet been well documented although there are many setups that use direction for extracting the ion flow. One possible reason for this is that many terrestrial experiments lack the ability to generate anisotropic electron distributions in the regions accessible by a Langmuir probe. Since the theory is lacking in understanding anisotropic distributions we will need to develop our own.

4.1.4 Our technique

Comparing all of the aforementioned options, we decided to use a directional Langmuir probe setup as we already have a fair number of the required materials, knowledge to interpret the signals, and may give the ability to extract the plasma flow and temperature anisotropy at the same time which would be interesting to see.

4.2 Hardware

The pressure anisotropy probe (PA probe) uses many of the same designs as the multi-tip Langmuir probe so we'll focus on highlighting any differences.

4.2.1 Shaft Components

The garages, shaft materials, and usage of centering collets is the same as described in 3.2.2. The only difference is the feed-thru.

4.2.1.1 Feed-thru

We use the same basic setup of two hermetically sealed DB-25s mounted to a KF-50 flange except for an addition of a through-hole that carries the Bdot signals. This is made by drilling a hole through the flange and then putting twelve wires along with a structural wire through the hole. It's vacuum sealed using an epoxy and mounting connection are soldered to both ends of the signal wires (an internal circuit board and an external DB-15) while heat shrink and teflon tubing is used to protect the signal wires.

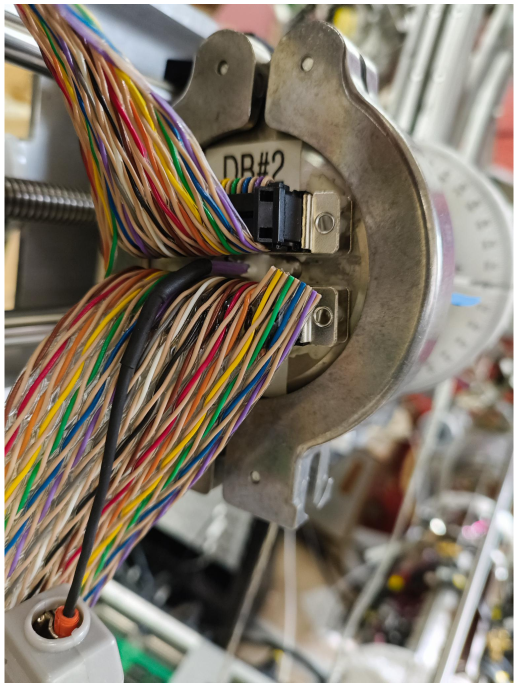


Figure 4.1: Picture of feed thru along with drawings of cross-sections.

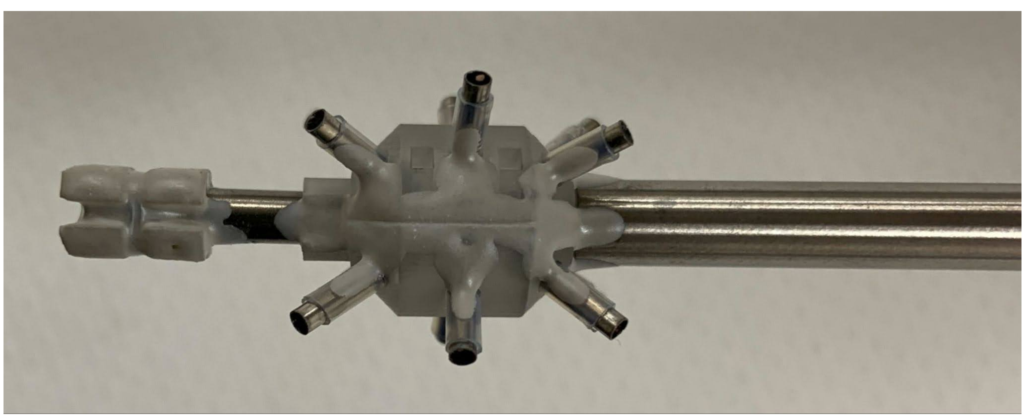


Figure 4.2: Picture of end of probe

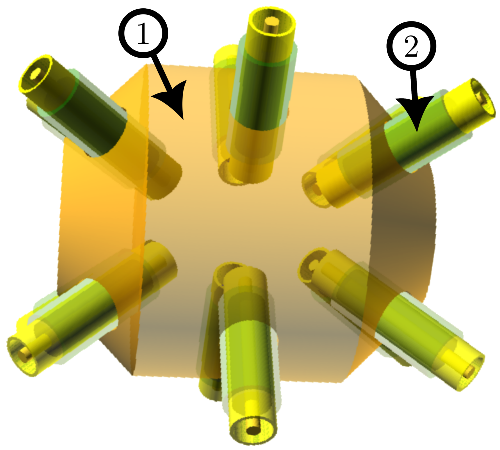


Figure 4.3: CAD diagram of full body

4.2.2 Tip Components

4.2.2.1 Probe body

Using the same techniques and considerations as 3.2.3.1 we've gone with a 3D printed probe body.

4.2.2.1.1 3D printed The body is printed using stereolithography on a printer in the physics department.

4.2.2.1.2 Two sides To make the probe building as simple as possible, the body is printed in two halves. This lets us construct each tip individually and then feed the tip and it's signal wires through the hole in the body half. Each half is press fit together after the tips are in place with some epoxy to keep them together and keep plasma out of the probe internals.

4.2.2.2 Tip material + size

As compared to the multitip probe, we've gone with stainless steel and copper as our conductors instead of molybdenum. Although both of these

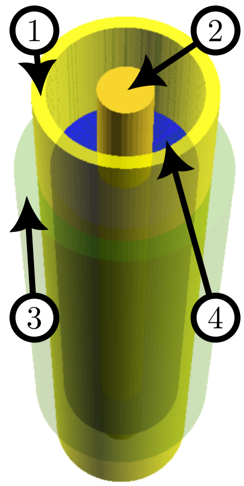


Figure 4.4: CAD diagram of probe tip

metals have higher secondary electron emission coefficients they are both still smaller at the low temperatures our plasmas attain Dekker (1958).

4.2.2.2.1 Shell The shell of the tip is made of 7 mm of stainless steel tubing. On one side, a wire is soldered for measuring the current to the shell and the other side is exposed to the plasma.

4.2.2.2.2 Core The core on the otherhand is just 24 AWG wire that is placed inside the shell with a limited amount of wire exposed from it's PTFE insulation. The other side of the wire is also soldered for measurment.

4.2.2.3 Tip Shielding

To restrict current to the tip to not overwhelm measurement and reduce tip heating, we shield both the shell and the core.

4.2.2.3.1 Shell Shielding The shell is shielded using teflon tubing that exposes some of the tip on the plasma facing side and helps to move the measurement region away from the body to avoid shadowing effects (see 3.5) as we want the shell to collect isotropically.

4.2.2.3.2 Core Shielding The core is shielded in two ways. The insulation on the core limits how much is exposed in total while the height of the core cylinder face relative to the shell edge limits the projected area the core sees as discussed in 4.4.1.3.3.

4.2.3 Bdot Coils

The design of the Bdot probe on the end of the PA probe is the same as that for the multitip Langmuir probe as discussed in 3.2.4.

4.2.4 Length Scales

The plasma parameters we must consider are the same as those for the multitip probe but we have a few other probe length scales specific to the PA probe which can all be seen in table 4.1.

4.3 Circuitry

Since the PA probe is also a Langmuir probe, it uses many of the same circuit systems as the multitip probe but custom built for this probe.

4.3.1 Bias Source

The biasing circuit for the PA probe is far simpler than that for the T_e probe as can be seen in figure 4.5. We only choose a single bias for each shot instead of splitting it between many biases. We do need to be able to change the bias between shots which occur every three minutes so we discharge the capacitor bank using a discharge resistor that is always connected.

Name	Symbol	Equation	Value
Core radius	r	—	
Shell inner radius	r	—	
Shell outer radius	r	—	
Exposed shell length	l	—	
Exposed core length	l	—	
Top of core to top of shell	l	—	
Poloidal angular separation	—	—	
Toroidal angular separation	—	—	
Across probe tip separation	—	—	
B-dot coil side length	—	—	3e-3
Debye length	λ_D	$\sqrt{\frac{\epsilon_0 T_e}{ne}}$	3e-5
Ion Gyroradius	r_i	$\frac{m_i\sqrt{2m_i\bar{v}_i}}{eB}$	6e-1
Electron Gyroradius	r_e	$\frac{m_e\sqrt{2m_e\bar{v}_e}}{eB}$	1.5e-1
Ion mean free path	λ_i	$\frac{\bar{v}_i}{\nu_{i,i}}$	9e-4
Electron mean free path	λ_e	$\frac{\bar{v}_e}{\nu_{e,e}}$	4.5
Electron energy relaxation length	λ_ε	* Eq. 5	4.5
Sheath thickness	h	$\sim \lambda_D$	3e-5

Table 4.1: Length scales of the PA probe. All lengths are in meters and are approximate. Refer to table 3.1 for plasma parameters and citations.

4.3.1.1 Principles of operation

The PA bias circuit system operates on the same principles as the multitip probe where we want a large charge reservoir to balance the current drawn by the probe and to keep the bias applied to the tip constant.

4.3.1.1.1 Charging Charging is done through V_{power}, a remotely operated power supply that can go up to 250 V, 375 mA. We charge the capacitors through a charging resistor which reduces the maximum current allowed to flow from the power supply which protects it and the capacitors.

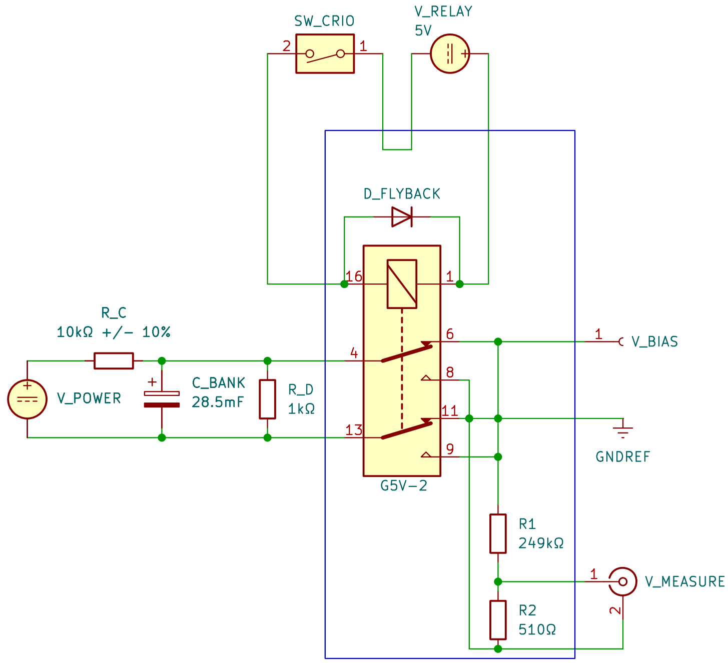


Figure 4.5: Biasing circuit for the PA probe. The charging resistor, R_c, let's us charge the capacitor bank without having a short circuit while the discharge resistor, R_d, let's us quickly change the bias on the capacitor bank when remotely changing the power supply voltage, V_b. Since the capacitors are electrolytic and we want both positive and negative biases, the relay system works to change what side of the capacitor bank is connected to ground. We also want to measure the voltage on the capacitor bank which is done after dividing the voltage down using R_1 and R_2.

4.3.1.1.2 Discharging To reduce the voltage to the tips quickly and for safety purposes we use a discharge resistor. This gives an RC time of approximately 30 seconds which, relative to a shot period of three minutes, is short enough to change the bias voltage to a new value between shots.

4.3.1.1.3 Swapping polarity We use electrolytic capacitors in our capacitor bank so we use a separate circuit that is connected to both sides of the capacitor bank. It consists of a relay that chooses which side of the capacitor is grounded and which side is connected to the probe. Thus

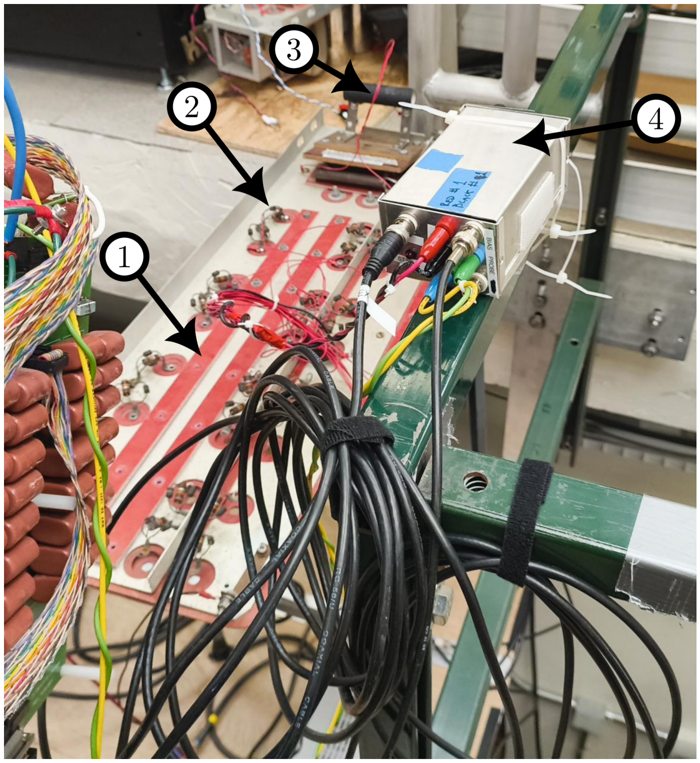


Figure 4.6

we can always charge the capacitors to a positive voltage and still get a negative output voltage.

4.3.1.2 How do we do this

Now that we've discussed the principles of how this thing should operate, let's dive into some of the details of really getting it built and working. An image of the system can be seen in figure 4.6.

4.3.1.2.1 Cannibalization A classic experimental technique is that of cannibalization; i.e. using parts from old systems that don't work quite like we want! In this case we've used a large capacitor bank, 1 in fig. 4.6, consisting of many individual electrolytic capacitors which are connected to plates for positive and negative bias. Between the positive bias plate and the positive terminal is a charging resistor and also a diode for capacitor protection, 2 in fig. 4.6.

4.3.1.2.2 Discharging We use a high power resistor, 3 in fig. 4.6, to discharge the capacitor bank without burning up the resistor.

4.3.1.2.3 Reducing voltage drop due to inductance Since we want to keep the bias voltage applied to the tip constant, we've taken a few steps to make that happen.

4.3.1.2.3.1 Twisting Wires One such solution is twisting all pairs of wires between the capacitor bank and the relay system and similarly between the relay and the board where the sense resistors are. Additionally instead of just having a single wire from the capacitors in the bank we've made it mutltistranded such that each capacitor has it's own wire to help reduce inductance.

4.3.1.2.3.2 Adding cap next to board We've also put a capacitor just before the sense resistor board that, while being low capacitance, helps keep the voltage constant by storing charge for biasing right where the probe samples the bias.

4.3.1.2.4 Reversing polarity We use a Double-Pole Double-Throw relay to select which side of the capacitor bank goes to ground. This is controlled remotely using a cRIO switch. Unfortunately we found that sometimes the relay breaks and if not caught quickly, leads to many wasted shots.

4.3.1.2.4.1 Why does the relay break We hypothesize that the relay breaks because of high transient current/voltage during switching. If the capacitor bank has some charge on it (say 25 V), there must be some current needed to shift the bias relative to ground because of parasitic capacitance with the lab. Additionally the capacitor we've placed at the sense resistor boards is a source of energy that could destroy the relay.

4.3.1.2.4.2 How do we fix this problem We've found that if we be careful about when we change from negative to positive bias leads to a longer relay lifetime. This is coded into the control interface such that there is a discharge delay which is dependent on the initial bias applied to the capacitor bank.

4.3.1.3 Calibration

Calibration of this power supply operates in the same way as that described in 3.3.1.2.3.

4.3.2 Probe Circuit

The probe circuit model can be seen in figure 4.7 which is reminiscent of 3.9. We've also hidden the where the digitizer measures to reduce clutter but to note, I_d is what the digitizer measures.

4.3.2.1 Single-ended measurement

There are a few different ways we could go about measuring the signals coming from the PA probe. One option is something like an oscilloscope which, while great at having a high sample rate, also has the major issue of grounding the signal through the measurement and the measurment is relative to the oscilloscope ground.

4.3.2.1.1 TREX cells These digitizers act similarly to an oscilloscope with many channels where the ground is shared between all measurement channels.

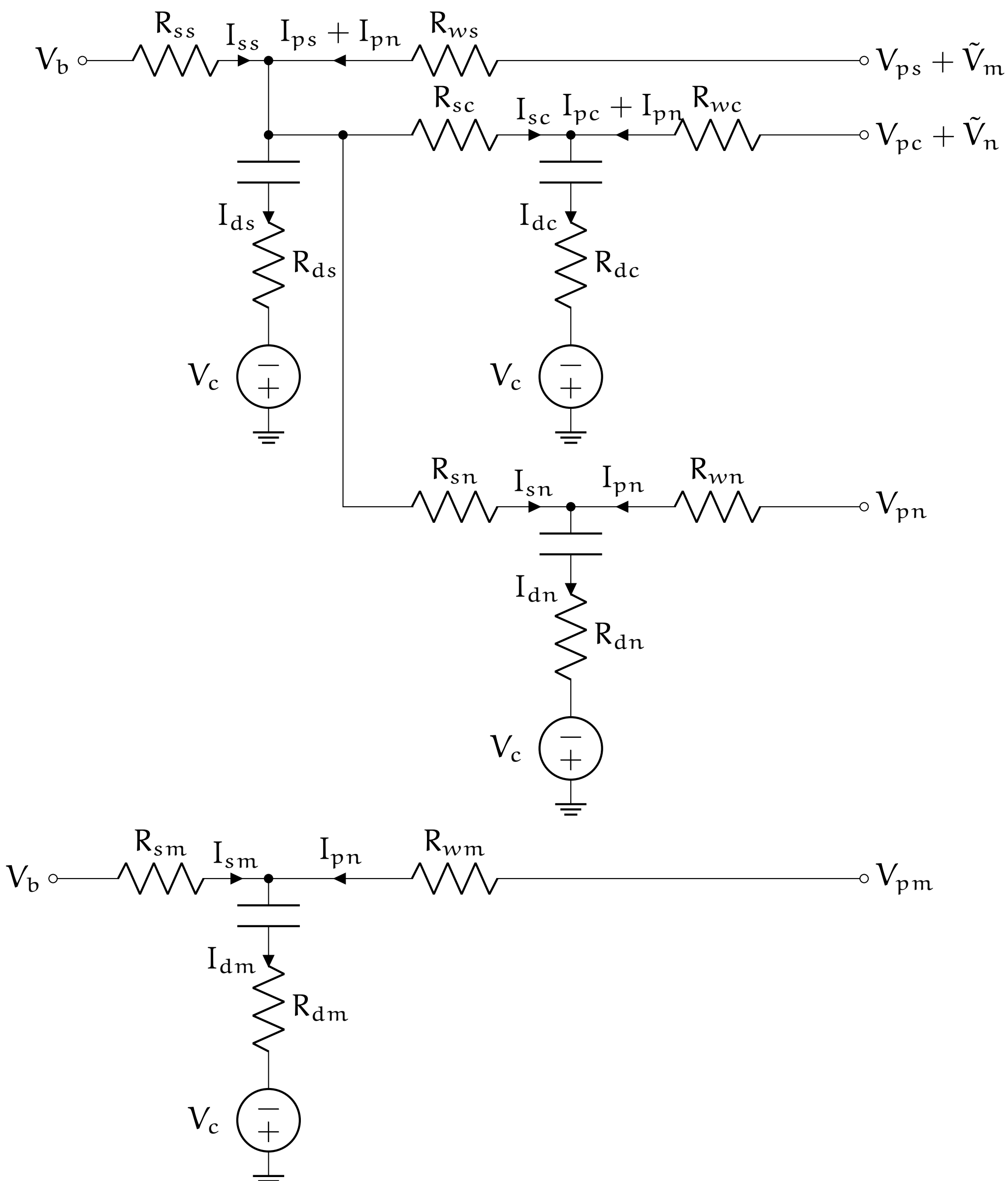


Figure 4.7: PA tip circuit model with non-static digitizer ground. Circuit for a single tip on the PA probe. The shell and core (subscripts s and c) both have thier own noises (m and n respectively). The digitizers, which measure the voltage across $R_{d\_}$, haven't been drawn to reduce clutter. This model assumes each digitizer is connected to changing ground that changes the same for all probes.

Table 4.2: Probe circuit component values

4.3.2.1.2 Problems Since the ground of the measurement is shared between all of the measurements, common mode noise very easily enters the measurement leading to wild oscillations when the capacitor bank for the drive coils triggers. Ideally subtracting off the noise measurement would correct for this but because of slight differences in circuit resistances and measurement gains, the subtraction does not sufficiently remove noise leaving an SNR of less than 0.1.

4.3.2.1.3 Next steps After a fair amount of time working to calibrate the circuit and remove noise, we decided to move onto use a different set of digitizers that use a differential measurement.

4.3.2.2 Differential measurement

Using the ACQ2X06s we can measure the difference between two signals instead of the difference between a signal and ground so we've changed the circuit to best account for that. The circuit components are the same as seen in 4.7 but with an additional parallel circuit.

4.3.2.2.1 Parallel circuitry Instead of having a single set of resistors and a high-pass capacitor for a tip, we've doubled the circuits on the sense resistor board as can be seen in the middle blue box in figure 4.8. For the shell noise and signal it's a simple case of just having another set that is parallel to the shell measurement. For the core, since it is connected to the shell circuit just prior to R_{sc} and R_{sn}, the parallel circuit is connected at the same point.

By using this parallel circuitry we drastically reduce the noise in the measurement which helps a lot.

4.3.2.3 Grounding Techniques

Good grounding leads to a happy diagnostic. Bad grounding leads to an unhappy graduate student.

Since we have immense amounts of current flowing when triggering the drive coils, capacitive and inductive coupling can occur leading to lots of noise in digitized measurements. A good review on grounding theory can be found in Horowitz and Hill (1989).

4.3.2.3.1 Types of grounds on the ACQ The ACQ2X06 digitizers (where $X \in \{1, 2\}$) have two types of ground. The first comes from the power cable plugged into the digitizer which we'll call the 'chassis' ground (as can be seen in figure 4.8). This ground is not directly used during measurement but is instead for safety in case of arcs. The second ground is that of the onboard differential amplifier (called the '0V' ground). This ground sets the limits of the amplifier and by tying the probe circuit ground to this ground, noise between the probe circuit and amplifier can be removed. The 'chassis' and '0V' grounds are connected by a high resistance so that the '0V' ground does not float away but also is decoupled.

4.3.2.3.2 Routing grounds Figuring out how to put all the grounds together was a long and arduous process. After much trial and error, we've come up with the solution found in 4.8. The entire system is powered by an isolation transformer, `T1`. This removes the building ground from the circuit and let's us optionally choose a different ground (we've neglected to attach a grounding wire at the transformer). Instead we ground the sytem through the 'chassis' ground of the digitizers and also at the sense resistor board to tie the base of the resistor dividers. The base of the resistor dividers is also directly connected to the '0V' ground which makes the divider measurements move with the opamp making the signal have low noise.

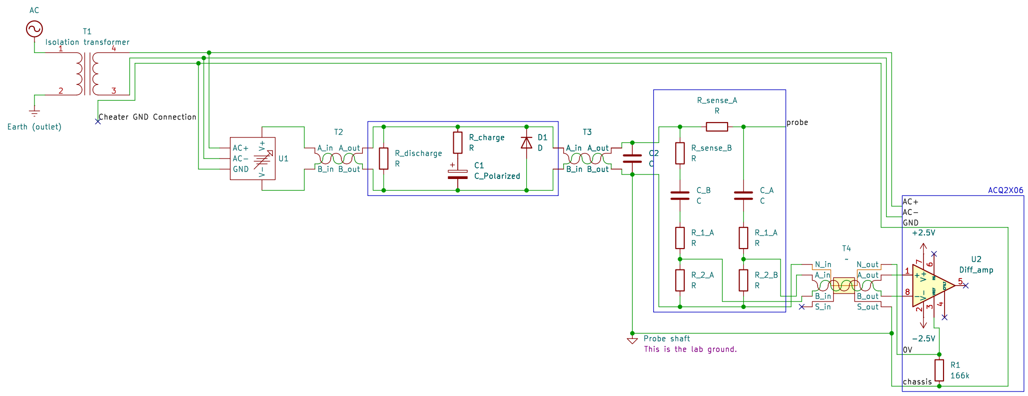


Figure 4.8: Schematic showing the ground paths for the PA probe. Power to the system comes from an isolation transformer which powers a controllable power supply, `U1`, and the digitizer, `ACQ2X06`. The bias capacitor bank, `C1`, is charged and that bias is applied to the probe board which contains the sense resiistors, high-pass capacitor, and dividers (`R_sense`, `C`, and (`R_1`, `R_2`)). Outputs from the board are then measured by the digitizer. The grounding wire is connected to the probe shaft which is then connected to the machine.

4.3.2.4 Swapping current paths

If our circuit components were ideal, each element would act the same as that same element for other tips. Since there is some variation in resistance between sense resistors and divider resistors, it's useful to have two current paths for the core measurements.

4.3.2.4.1 Why do we want to do this The current collected by the core is quite small and the raw voltage measurement, V_{dc}, is dominated by the shell current by a factor of around 20. Now R_{sc} (R_{sn}) and R_{dc} (R_{dn}) make a resistor divider that divides down the core signal (noise) measurement and then the core signal and noise measurements, V_{dc} and V_{dn}, are subtracted to get the core signal.

4.3.2.4.1.1 Deviations in resistance If the resistors in the circuit are different though, the subtraction does not fully remove the shell signal. This causes the derived core voltage and current to be either anomalously high or low compared to it's actual value.

4.3.2.4.1.2 Correcting for this The reason the shell leaks into the core measurement is because of the different divider ratios between the core signal and noise. If we swap the resistors for those two signals though, we can see the effect of both adding shell signal and subtracting shell signal. We can then average the signal between the original path and the swapped path to remove the shell contribution.

4.3.2.4.2 Circuitry to do this The circuitry we use can be seen in figure 4.9 and the corresponding schematic in figure 4.10. The circuit has a number of components used for isolating the CRIO from the circuit and also keeping current through the CRIO to a minimum.

4.3.2.4.2.1 Only swap core signal and noise As we are trying to remove the influence of the different core noise and shell circuits, we only want to swap those two lines. The shell signal and noise are minimally affected by the core and the small resistor differences matter little.

4.3.2.4.2.2 Many relays Each tip needs it's core measurements swapped so in total we have 12 relays. This means that we require a sufficient amount of current for all these relays and thus want to isolate the switching system from the CRIO which is low power.

4.3.2.4.2.3 Remote control To control this switching system, we isolate the CRIO using an optoisolator and use a switch in the CRIO that we can control via LabView. Since the signals we are switching are

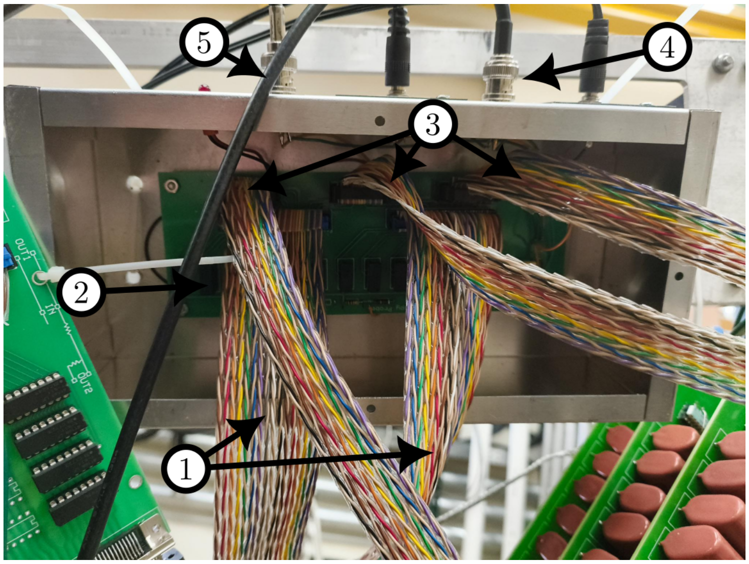


Figure 4.9: Image of swapping box

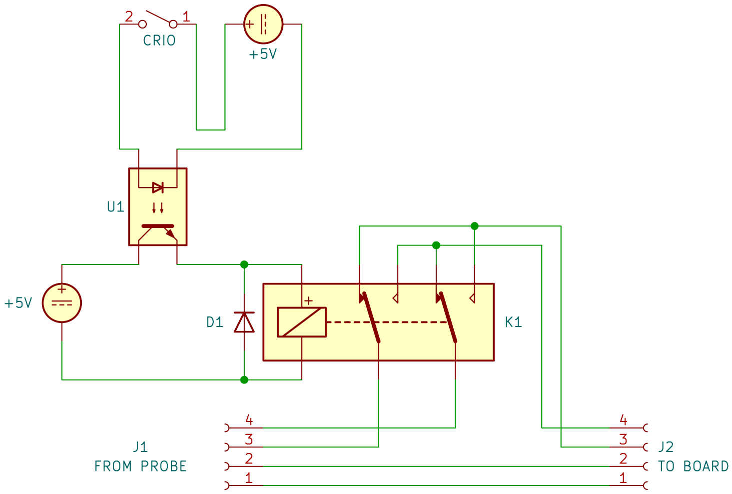


Figure 4.10: Diagram of swapping

connected to the bias capacitor bank, we want to keep these high voltage high capacitance lines far away from the expensive CRIO components.

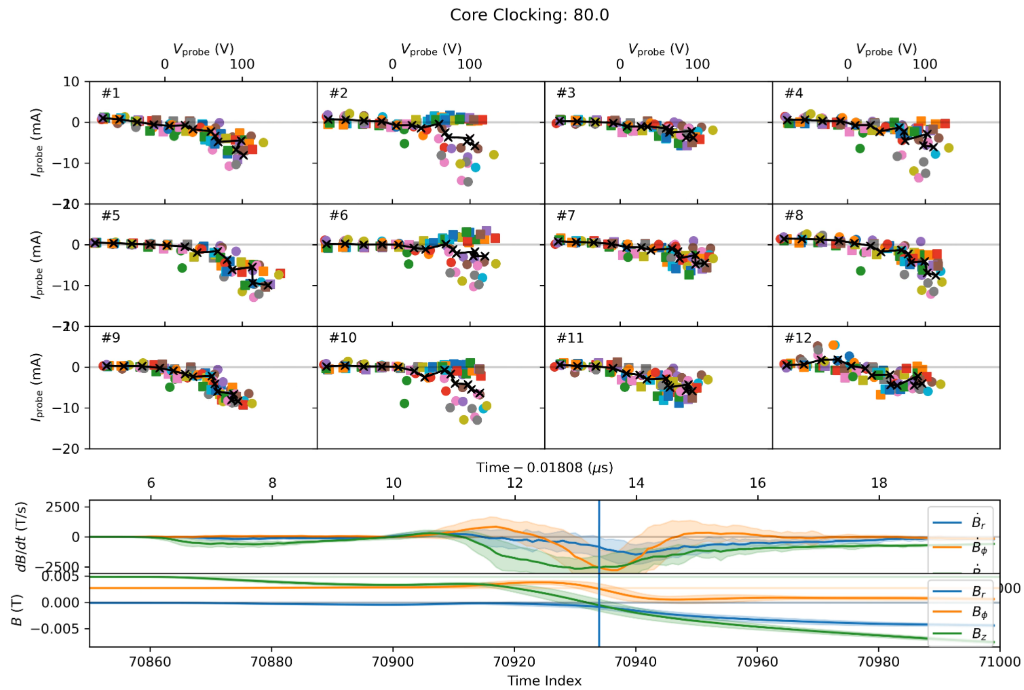


Figure 4.11: Time frame for core movie of IV with separation between relay

4.3.2.4.3 How does this change results

Figure 4.11 shows the difference between the unswapped (circles) and swapped (squares) core measurements. It is especially clear in tips 2, 6, and 10 where the unswapped measurements are far higher than the swapped because the shell is either adding or subtracting from the signal. By averaging these two options, as described in 4.5.1.3, we can correct for this problem.

4.3.2.5 Circuit Solution

The circuit for the PA probe tips, figure 4.7, acts similarly as that of the T_e probe, figure 3.10. The bias is input as V_b and passes through a sense resistor which acts to bring the probe bias closer to the plasma floating potential (where the current from the plasma to the probe is 0). The core of each tip is attached to the shell circuit because we want the tip core to have a similar bias as the shell. Since the core is smaller and sees less plasma it would not follow the plasma floating potential as much without that connection.

4.3.2.5.1 Solution Much of this analysis is the same as that done for the multitip Langmuir probe in 3.3.2.4 so we will gloss over a large portion and jump right into the final equations for the quantities we want to find, V_{ps}, V_{pc}, I_{ps}, and I_{pc}.

$$
\begin{aligned}
I_{ps} =& -\frac{I_{dc}R_{dc}}{R_{sc}} + I_{dm}\left(-\frac{R_{dm}}{R_{ss}} - \frac{R_{sm}}{R_{ss}}\right) + I_{dn}\left(\frac{R_{dn}R_{sm}}{R_{sn}R_{ss}} - \frac{2R_{dn}}{R_{sn}} + \frac{R_{sm}}{R_{ss}} - 1\right) \\
&+ I_{ds}\left(-\frac{R_{ds}R_{sm}}{R_{sn}R_{ss}} + \frac{R_{ds}}{R_{ss}} + \frac{2R_{ds}}{R_{sn}} + \frac{R_{ds}}{R_{sc}} + 1\right) \\
V_{ps} =& -\frac{I_{dc}R_{dc}R_{ws}}{R_{sc}} + I_{dm}\left(-R_{dm} - \frac{R_{dm}R_{ws}}{R_{ss}} - R_{sm} - \frac{R_{sm}R_{ws}}{R_{ss}}\right) \\
&+ I_{dn}\left(\frac{R_{dn}R_{sm}}{R_{sn}} + \frac{R_{dn}R_{sm}R_{ws}}{R_{sn}R_{ss}} - \frac{R_{dn}R_{ws}}{R_{sn}} + R_{sm} + \frac{R_{sm}R_{ws}}{R_{ss}}\right) \\
&+ I_{ds}\left(-\frac{R_{ds}R_{sm}}{R_{sn}} - \frac{R_{ds}R_{sm}R_{ws}}{R_{sn}R_{ss}} + R_{ds} + \frac{R_{ds}R_{ws}}{R_{ss}} + \frac{R_{ds}R_{ws}}{R_{sn}} + \frac{R_{ds}R_{ws}}{R_{sc}} + R_{ws}\right) \\
I_{pc} =& \frac{I_{dc}R_{sn}\left(R_{dc} + R_{sc}\right) - I_{dn}R_{sc}\left(R_{dn} + R_{sn}\right) + I_{ds}R_{ds}\left(R_{sc} - R_{sn}\right)}{R_{sc}R_{sn}} \\
V_{pc} =& I_{dc}\left(R_{dc} + \frac{R_{dc}R_{wc}}{R_{sc}} + R_{wc}\right) + I_{dm}\left(-R_{dm} - R_{sm}\right) + I_{dn}\left(\frac{R_{dn}R_{sm}}{R_{sn}} + R_{sm}\right) \\
&+ I_{ds}\left(-\frac{R_{ds}R_{sm}}{R_{sn}} - \frac{R_{ds}R_{wc}}{R_{sc}}\right)
\end{aligned}
$$

4.3.2.5.2 Assumptions Here we have the same assumption of noise coming from a changing measurement ground but we now can account for a second source of noise which is extraneous current flowing down the probe wires due to coupling or other strange effects. We've also assumed, similar to previously, that the high pass capacitors change too slowly during a shot for their charge to matter.

4.3.2.6 Calibration

Calibration of the PA probe is once again done very similarly to that of the multitip Langmuir probe, 3.3.2.5. Here we'll discuss some of the differences.

4.3.2.6.1 Calibrating in Ion Saturation During the initial plasma equilibrium, the plasma is quite isotropic and along with the isotropic collection of the shells, we can calibrate between each shell the relative areas.

4.3.2.6.2 Calibrating on the exponential Also during that equilibrium, the plasma temperature is quite low (around 4 eV) so by biasing the shells such that they lie near the plasma potential on the exponential, they should all have nearly the same probe potential which lets us calibrate the wire resistance.

4.3.2.6.3 Rotation calibration Although the initial equilibrium is quite isotropic, there is some flow from the plasma guns and cross-field diffusion of the plasma to higher radii. Thus to correct for these issues we calibrate the same core parameters as the shell by rotating the probe between shots such that each direction is covered by different tips. Then we can use the different shots with tips in the same direction in both the ion saturation region and on the exponential to get the core wire resistance and area.

4.3.2.7 Capacitive coupling

This probe was built using a technique that made sense in the moment but leads to some effects that we need to remove after data collection. To make the noise look as similar as possible to the signal, we used fine twisted pair wire for the shell signal and noise and similarly for the core signal and noise. This leads to a capacitive coupling between the signal and noise measurements due to a difference in voltage between the two wires.

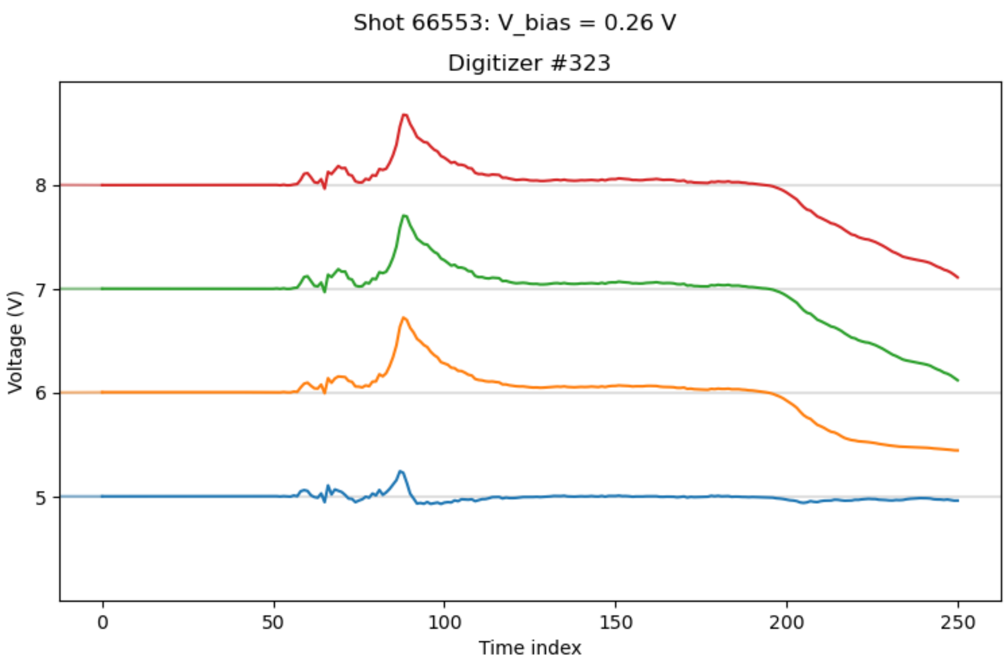


Figure 4.12: Example of the raw signal measured by the digitizers. This signal is from tip #2. Each signal has it's zero at the corresponding channel of the digitizer (i.e. the signal that starts at 5 V is really starting at 0 V but is offset by the channel number). Channel 5 is the shell noise wire, channel 6 is the shell signal wire, channel 7 is the core noise wire, and channel 8 is the core signal wire.

4.3.2.7.1 What do we see As can be seen in figure 4.12, the shell noise (signal number 5) is approximately the derivative of the shell current (signal number 6) multiplied by some constant. This is the case for all tips but the coefficient multiplying the derivative can be different. The same effect should also occur for the core measurements but are more difficult to see since they are dominated by the shell signal.

4.3.2.7.2 Where does it come from The combination of sense resistors and closely wound wires is a recipe for a RC circuit-like effects. Intuitively, since the signal looks like the derivative let's look at what a circuit

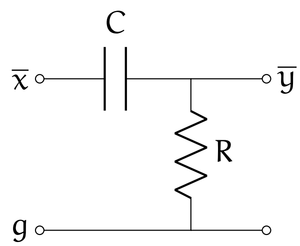


Figure 4.13: A passive differentiator. This figure is reproduced from Wikipedia.

differentiator is.

4.3.2.7.2.1 A differentiator As can be seen in figure 4.13, a passive differentiator is just a resistor and capacitor acting like a divider. When $\overline{x} - g$ varies, at low frequency the capacitor acts as a break in the circuit leading to no voltage difference on $\overline{y}$ but at very high frequencies all the voltage is transfered to $\overline{y}$ since the capacitor acts as a short. The transfer function for this circuit can be written as

$$\overline{Y} = \frac{sRC}{1 + sRC}\overline{X} \tag{4.1}$$

where $\overline{X} = \overline{x} - g$ and $\overline{Y} = \overline{y} - g$ and $\overline{x}$, $\overline{y}$, and g are the voltages at their respective nodes.

4.3.2.7.2.2 High-pass capacitor and resistor divider This differentiator looks very similar to the high pass capacitors and resistor dividers we use in the circuit but because of the high resistance and capacitance they have an RC time greater than a millisecond. Thus we can ignore the differentiator effects there.

4.3.2.7.2.3 Shell/core circuit Let's start by just examining the shell but the core undergoes a similar method. Figure 4.14 is the same as figure

4.7 except that we've added the capacitive coupling as a capacitor, C_c. We've also removed the core circuit as it does not play a significant role in the shell analysis. In the case of the core analysis, $V_b \rightarrow V_s + V_b$, $V_s \rightarrow V_c$, and $V_m \rightarrow V_n$. As can be seen in the simplified circuit at the bottom of figure 4.14, looking at everything to the left of C_c, the shell voltage measurement (V_s) is just the divided down $V_p - V_b$. Now instead if we look at everything on the right of R_p and $V_p - V_b$ (including those two elements) we see that we have a differentiator and also a resistor divider. The R_w in this right circuit is difficult to calculate as capacitive coupling occurs along the entire wire so it's simpler to just ignore R_w everywhere as it plays a small part in the analysis.

4.3.2.7.3 Magnitude of effect We can calculate the capacitive coupling using the wire parameters. We are using very fine twisted pair wire with a wire diameter of 0.003 inches, a separation between wire centers of 0.003 inches, and the dielectric constant of PTFE is 2.2 (this is the enamel coating material on the wires). We then get a capacitance per meter of 84.2 pF / m. Using a sense resistance of 680 Ω we find that $RC \approx 0.11$ μs. Since our signals are on the order of 0.1 μs ($s \approxeq 6 \cdot 10^7$ rad/s) the magnitude of the coefficient in 4.1 is ≈ 0.8. The capacitive coupling is a bit over estimated since the wires are not perfectly twisted but this is good enough to let us know that this is an effect to remove.

4.3.2.7.4 Removing effect As of this writing, the best way to take care of this effect is to ignore it. This is because the only reason we may want to remove this effect is if we want to extract the common mode noise to then remove from the signal. For the shell, the magnitude of the coupling on the noise signal can be up to 1/3rd of the shell signal whereas the common mode noise for the shell is far smaller (maybe around 1%). For the core, since both the core signal and core noise have a very similar voltage, the capacitive coupling is very small and although the common mode noise

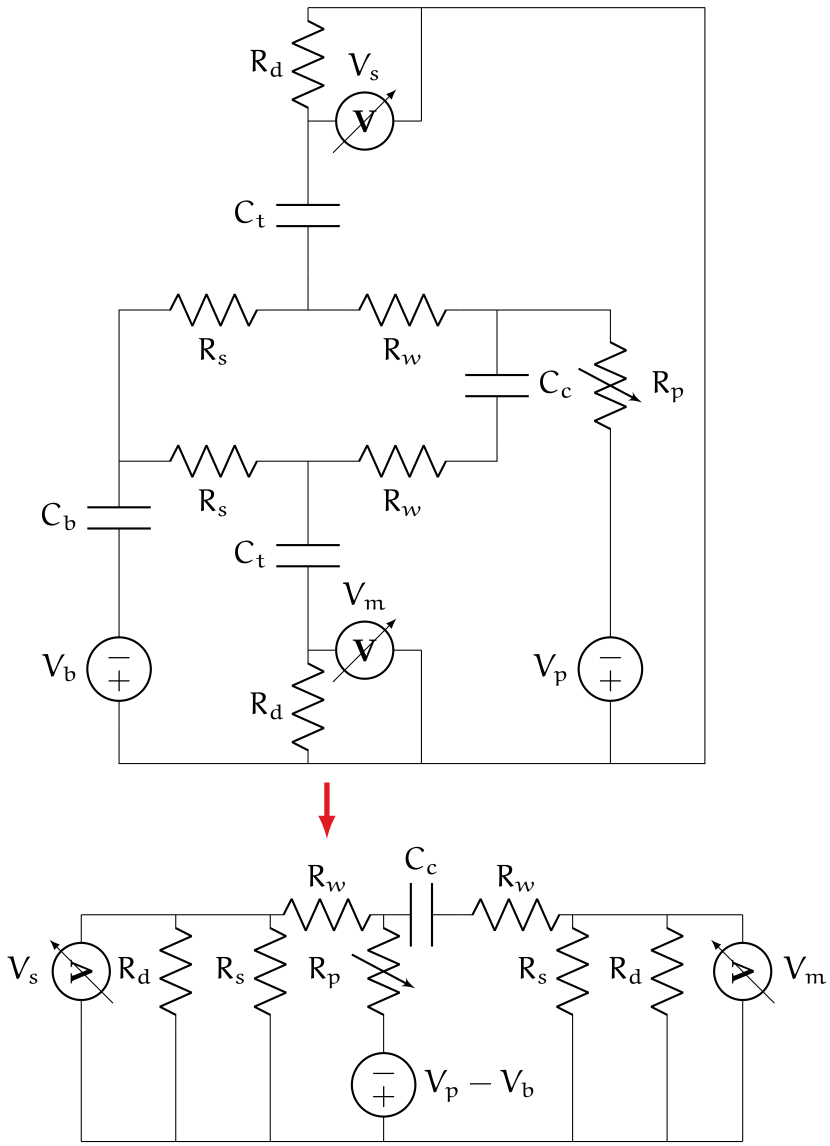


Figure 4.14: The capacitive coupling of the shell signal and the shell noise which are coupled by capacitor C_c. The signals are measured via V_n and V_s. The left side of the top circuit is the bias voltage (V_b) applied to the biasing capacitor (C_b). R_p is the resistance of the plasma which is calculated from Ohm's law: $V_{probe} - V_{plasma} = I_{probe}R_{plasma}$. The p's in V_p and R_p are for 'plasma'. To achieve the bottom circuit, we've simplified the top circuit by combining voltage supplies and removing capacitances that are too large to play into the capacitive coupling effect we see.

is larger the capacitive coupling is very small. We have done some work extracting transfer functions to account for this effect but this is ongoing.

4.4 Plasma Theory

4.4.1 A full model

4.4.1.1 Plasma Distribution

We assume that we have a drifting bi-Maxwellian for the electrons where the temperature parallel to the magnetic field, **B**, is different than the temperature perpendicular to it. If we let the drift velocity be $\mathbf{v_D}$ then we can write our distrubution function as the following:

$$f_e(\mathbf{v}) = \left(\frac{m_e}{2\pi k}\right)^{3/2} \frac{n}{\sqrt{T_{e\perp}^2 T_{e\parallel}}} \exp\left[-\frac{em_e}{2}\left(\frac{(\mathbf{v}-\mathbf{v_D})_\parallel^2}{T_{e\parallel}} + \frac{(\mathbf{v}-\mathbf{v_D})_\perp^2}{T_{e\perp}}\right)\right]. \tag{4.2}$$

We can extract the parallel and perpendicular components since

$$\begin{aligned}
(\mathbf{v}-\mathbf{v_D})_\parallel^2 &= \left((\mathbf{v}-\mathbf{v_D})\cdot\hat{b}\right)^2 \\
(\mathbf{v}-\mathbf{v_D})_\perp^2 &= \left((\mathbf{v}-\mathbf{v_D}) - \left((\mathbf{v}-\mathbf{v_D})\cdot\hat{b}\right)\hat{b}\right)^2
\end{aligned}$$

For the ions we'll assume that they are isotropic in temperature as most work with langmuir probes has shown that ion temperature effects are not highly impactful. Another way to say this is it's very difficult to extract the ion temperature from IV curves because it plays a small effect. We still assume that the ions drift with the electrons and often the drift speed is above the ion sound speed and thus our probe acts like a mach probe in many ways.

4.4.1.2 Ion particle flux

For the attracted ions we are fortunate. The core of each tip recieves very little ion current throughout the shot so having high accuracy is not necessary. Thus we'll use a relatively simple model by Hutchinson derived from a set of PIC simulations Hutchinson (2002a). Although this model doesn't work for large Debye length, we'll continue our approximation of small Debye length.

4.4.1.3 Electron particle flux

We split the calculation of the current to the probe into a multistage process. We will only model the repelled electrons using 4.2 as modeling the attracted electrons is a much more difficult task.

To calculate the total electron current to the core we need to calculate the flux to the probe from a specific direction. We'll assume that we can use energy conservation to get the minimum velocity for an electron to hit the probe and we only care about the electron velocity perpendicular to the probe surface. Once we integrate up the distribution function for a specific direction and minimum energy then we can integrate over all angles while multiplying by the area of the probe to get the electron current to the probe.

4.4.1.3.1 Coordinate System Let's orient our velocity space coordinate system along the probe direction. The position coordinate system is ignorable as we assume that the small separation in tip position is ignorable. Let v be the radial magnitude, θ_T be the polar angle, and φ_T be the azimuthal angle. $\theta_T = 0$ along the tip direction pointed into the plasma and away from the probe body.

Let $\hat{t}_1$, $\hat{t}_2$, and $\hat{t}_3$ be the cartesian unit vectors in this coordinate system. The third unit vector is pointed along $\theta_T = 0$. $\hat{t}_1$ and $\hat{t}_2$ are arbitrarily chosen to be perpendicular to $\hat{t}_3$. We can get the direction of the tip from

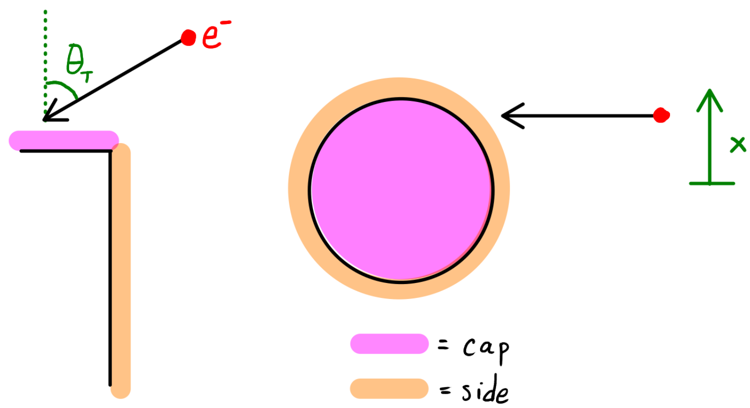


Figure 4.15: Diagram of electron coming from a direction θ_T and finite impact parameter on the side of the core.

how the probe is attached to it's port on the surface of the machine. To convert a vector, $\mathbf{u}_{mach}$, in machine cartesian coordinates into a vector in this new coordinate system we take $u_i = \mathbf{u}_{mach} \cdot \hat{t}_i$. This gives us the components of $\mathbf{u}_T$ where the T subscript means the tip coordinates.

4.4.1.3.2 Minimum velocity Let's orient our coordinate system such that $\theta_T = 0$ in spherical coordinates is along the axis of the tip as can be seen in figure 4.15. Then the perpendicular velocity is given by

$$v_{\perp,c} = v\cos(\theta_T)$$
$$v_{\perp,s} = v\sin(\theta_T)\sqrt{1 - \frac{x^2}{r^2}}$$

Since the electron needs to overcome the probe potential this means

$$
\begin{aligned}
|v_\perp| &\geqslant v_\perp^{\min}(x) \\
\frac{1}{2} m_e v_\perp^{\min}(x)^2 &= e(V_p - V_0) \\
\Rightarrow v_\perp^{\min}(x) &= \sqrt{\frac{2e}{m_e}(V_p - V_0)} \\
\Rightarrow v_c^{\min}(x) &= \frac{1}{\cos\theta_T}\sqrt{\frac{2e}{m_e}(V_p - V_0)} \\
\Rightarrow v_s^{\min}(x) &= \frac{1}{\sin(\theta_T)\sqrt{1 - x^2/r^2}}\sqrt{\frac{2e}{m_e}(V_p - V_0)}
\end{aligned}
$$

where V_0 is the plasma potential far from the probe.

4.4.1.3.3 Area Using the same coordinate system we can calculate the area of the cap, c, and side, s, of the core. What we want is the area projected along some direction as electrons from that direction will only see that limited amount of area. The easiest way to do this is to do an integral from $x = -r$ to $x = +r$ where r is the radius of the core. At any value of x, we want to find the height of exposed tip. As θ_T increases, the outer shell will cover the side of the core and then the cap. The equations for the projected height are

$$
\begin{aligned}
h_{\text{bottom},s}(x) &= -\cos(\theta_T)\sqrt{r^2 - x^2} \\
h_{\text{top},s}(x) &= -\cos(\theta_T)\sqrt{r^2 - x^2} + h\sin(\theta_T) \\
h_{\text{bottom},c}(x) &= -\cos(\theta_T)\sqrt{r^2 - x^2} + h\sin(\theta_T) \\
h_{\text{top},c}(x) &= \cos(\theta_T)\sqrt{r^2 - x^2} + h\sin(\theta_T) \\
h_{\text{top,shell}}(x) &= -\cos(\theta_T)\sqrt{R^2 - x^2} + H\sin(\theta_T)
\end{aligned}
$$

where R is the radius of the shell. The constants h and H are the heights of the core and shell respectively measured from the insulation height.

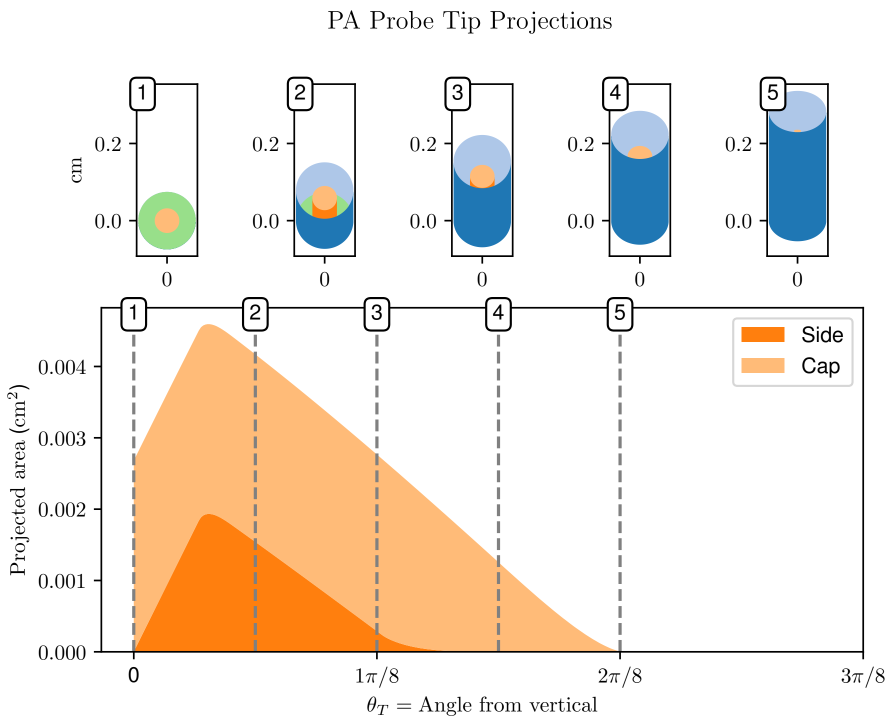


Figure 4.16: The bottom plot shows the projected area of the core as a function of direction. The top plots are what the probe tip looks like from specific direction associated with the vertical dashed grey lines. The shell is blue, insulation green, and core orange.

The total height for the side and cap including the covering due to the shell is

$$h_s(x) = h_{top,s}(x) - \min\left[h_{top,s}(x), \max\left[h_{bottom,s}(x), h_{top,shell}(x)\right]\right]$$
$$h_c(x) = h_{top,c}(x) - \min\left[h_{top,c}(x), \max\left[h_{bottom,c}(x), h_{top,shell}(x)\right]\right]$$

There is a maximum angle at which the side of the core gets fully covered by the shell. There is a larger angle similarly associated with the cap. The conditions of coverage are

$$
\begin{aligned}
h_{\text{top,shell}}(x = r, \theta_T = \theta_{T,s}^{max}) &= h_{\text{top},s}(x = r, \theta_T = \theta_{T,s}^{max}) \quad \text{Side} \\
\Rightarrow \theta_{T,s}^{max} &= \tan^{-1}\left(\frac{\sqrt{R^2 - r^2}}{H - h}\right) \\
h_{\text{top,shell}}(x = 0, \theta_T = \theta_{T,c}^{max}) &= h_{\text{top},c}(x = 0, \theta_T = \theta_{T,c}^{max}) \quad \text{Cap} \\
\Rightarrow \theta_{T,c}^{max} &= \tan^{-1}\left(\frac{R + r}{H - h}\right)
\end{aligned}
$$

4.4.1.3.4 Particle Flux We are now ready to calculate the particle flux to the probe. We'll integrate the distribution function along a specific φ_T and θ_T. Then we'll do a solid angle integral to get the total flux. The particle flux is calculated by multiplying the area along a projected direction by the number of particles from that direction alongwith the velocity perpendicular to the probe.

$$
\Gamma_e = \int_0^{2\pi} d\varphi_T \sum_{l \in \{s,e\}} \int_0^{\theta_{T,l}^{max}} d\theta_T \int_{-r}^{r} dx \int_{v_l^{min}(\theta_T, x)}^{\infty} dv\, v^2 \sin(\theta_T) h_l(x) v_{\perp,l} f_e(v, \theta_T, \varphi_T) \tag{4.3}
$$

The $v^2 \sin(\theta_T)$ is due to spherical coordinates. $h_l(x)$ is the height of the projected area going from left to right in the small plots of 4.16. The $v_{\perp,l}$ term accounts for the velocity perpendicular to the probe and f_e is the distribution function.

We will remove the integral over x in order to simplify the problem. Since the side of the probe has much less projected area and is more affected by non-planar effects this approximation should have only a small effect. The particle flux to the side and core can be seen in fig 4.17. It can be seen that the flux to the side of the core is very small relative to the cap flux and only starts to become significant near $V_p = 0$.

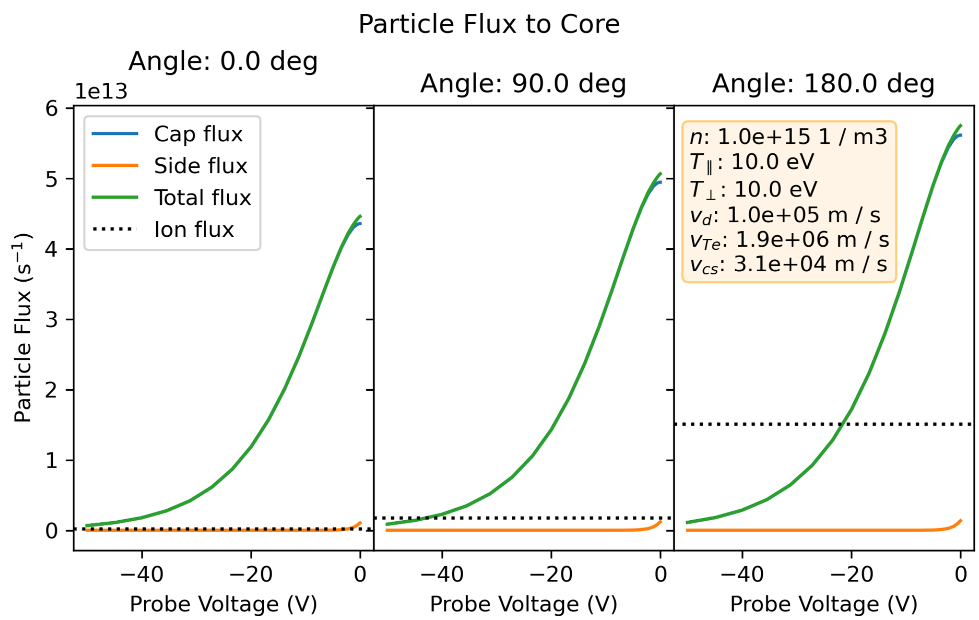


Figure 4.17: Example particle flux to the core for the electrons and ions of a flowing Maxwellian. The ion particle flux is much less for the tip pointed away from the flow (Angle = 0) compared to the tip pointed into the flow (Angle = 180). The electron particle flux is much less affected by the flow. Also, the particle flux to the side of the tip is very small in comparison to the flux to the cap.

4.4.1.4 Theoretical JV curves

Now that we have developed a theory for the expected current to a probe given some plasma parameters, let's take a look at what we expect the measurements to be given certain plasma parameters.

4.4.1.4.1 Reading radar plots Normally, IV-characteristics are read using a 2D plot of the probe potential vs. probe current (or current density). As we want to incorporate a directional aspect to this plot let's use a radar plot.

4.4.1.4.1.1 Radar axis The radar plot has a direction component which is the angle around the circle. For the probe potential we use the radial distance and choose a minimum voltage to be at zero radius (not plotting any points with bias less that the minimum voltage). The color is then the current to the probe at a given angle and voltage. We also draw a grey line where the floating potential is (i.e. there is no current to the probe).

4.4.1.4.2 Solutions Figure 4.18 shows some example electron distribution functions (top row) and their corresponding IV-characteristic radar plots (bottom row). In a Maxwellian plasma with no drift (left column) the IV radar plot looks the same for any angle. This is as expected since the plasma has no directional dependence. Once we add a drift, middle column, the distribution changes shape. In this case the plasma is flowing upward with a drift velocity that is small relative to the electron thermal speed but is on the order of tenths of ion sound speed. This means that the mach effect will come into play which leads to more ion current hitting probes pointing into the flow (pointing downwards) and very little current hitting probes pointed in the direction of flow. Finally, we can give a drifting bi-Maxwellian which will have both a Mach effect and a difference in temperature. Here we've set the drift to be upwards and the magnetic field direction (parallel direction) to be up and to the left. When this is the case we have two interesting features. First, we get the same ion current directional difference as seen for the drifting Maxwellian. Secondly, and more revealing, the electron current to the probe is much higher in the direction of higher temperature than in the direction of low temperature. This is as to be expected because there is higher electron flux in the high electron temperature direction.

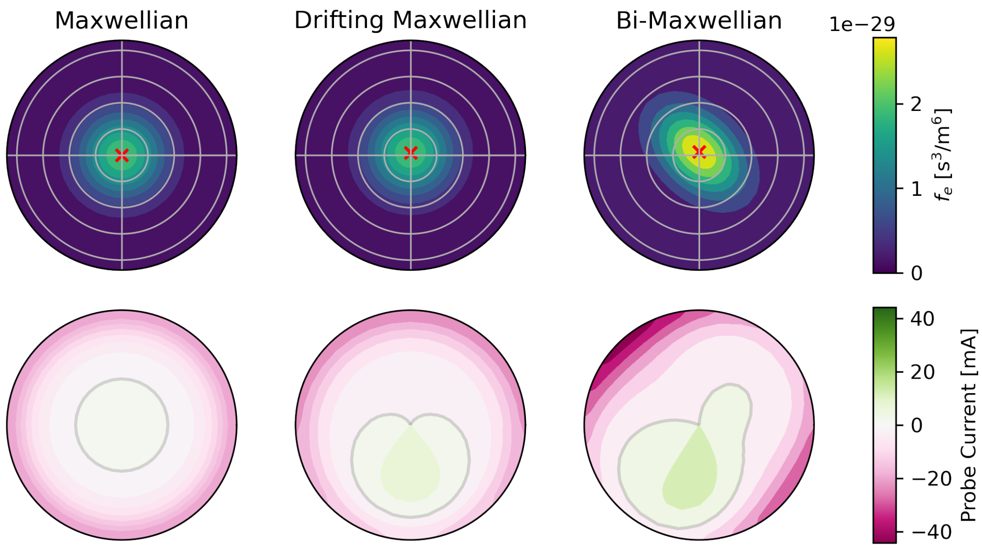


Figure 4.18: Figure of theoretical plasma distributions and JV curves

4.4.2 Individual tip fitting

As an alternative to the full model we just described, let's look into another option for extracting the plasma distribution. Instead of fitting all the tips at once, we can try fitting each using a routine similar to that for the multitip Langmuir probe and then combining each of those individual fits.

4.4.2.1 Why do this instead

We've found that the full model suffers from some issues in regards to finding a good fit to the measured data. One problem is that it is computationally intensive to recalculate the IV-characteristic when optimizing the parameters. Additionally, we are fitting six parameters at once! Namely, the electron density, parallel temperature, perpendicular temperature, and three components of the drift velocity. Even ignoring the drift velocity still leads to exceedingly long optimization times that don't fit the data all that well. One other reason that this may be the case is there may exist non-Maxwellian features in the plasma and the simplicity of this model may not correctly reproduce the actual plasma response to an acceptable degree.

4.4.2.2 Individual Tip Theories

Let's take a look as to the options we have available for fitting each tip separately.

4.4.2.2.1 Multitip methods In sections 3.4.1 and 3.4.2 we discussed two plasma theories to predict the expected current to a simple Langmuir probe given a set of plasma parameters describing an unmagnetized Maxwellian plasma. Both of these methods, and many other Langmuir probe theories, show the same general trends for the expected current to a Langmuir probe independent of theory approximations and magnetization. Thus one thing we can try doing is applying these theories to the PA probe in which we attempt to fit each tip using the code architecture as described in 3.5.2.

4.4.2.2.2 Exponential While both of the previous theories, Hutchinsons and BRL, work well for a normal Langmuir probe, we find that the geometry of our system may lead to different ratios of the electron to ion current than expected from one of these physically derived theories. Why might that be?

4.4.2.2.2.1 Ions don't hit core Figure 4.19 shows an approximate sheath around the core (orange rectangle) and shell (blue rectangles) as a cross section of the probe tip. Ions will be consumed by the shell and electrons will be focused towards the core leading to a lower than expected ratio of ion versus electron current to the core.

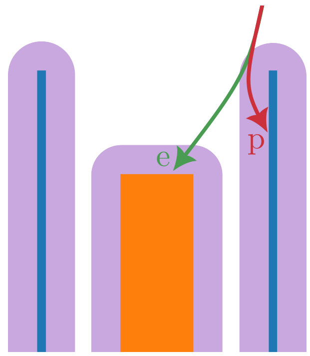


Figure 4.19: This figure shows some what may occur to ions in comparison to electrons when the probe is attracting ions. Ions, in this case protons signified by 'p' in red, are attracted to the probe and when passing close to the shell on their way to the core they hit the shell. The electrons (green, e) are instead repelled from the shell (blue rectangles) and focused toward the core (orange rectangle). The purple regions are the extent of the sheath over which there is a non-negligible electric field pointed toward the conductors.

4.4.2.2.2.2 More Free Parameters As already noted, the current to a Langmuir probe is well approximated by a constant ion current and an exponential electron current like as seen in Hutchinsons model. The ratio of these currents is set by physical assumptions and derivations that may not hold for our system so we choose an additional free parameter such that we are fitting the electron density, electron temperature, plasma potential, and the ion density. Although quasineutrality should hold in the plasma the separation between densities allows us to better model mach effects and take care of shell ion consumption.

4.4.2.2.2.3 Model equation We could fit the all the plasma parameters we just discussed but for increased fitting speed and simplicity, we've taken our model current to be

$$I_p(V_p) = -A\exp(V_p/B) + C. \tag{4.4}$$

All the parameters to fit $(A, B, \text{and } C)$ are greater than zero.

4.4.2.2.2.4 Plasma parameters to exponential parameters Using Hutchinsons model, equation 3.17, we can write the terms in the exponential model, eq. 4.4, as

$$A(n_e, T_e; V_{\text{plasma}}) = n_e e \sqrt{\frac{eT_e}{m_i}} \left[\frac{1}{2} \sqrt{\frac{2m_i}{\pi m_e}} \exp\left(\frac{-V_{\text{plasma}}}{T_e} \right) \right] \tag{4.5}$$

$$B(T_e) = T_e \tag{4.6}$$

$$C(n_i, T_e; V_{\text{plasma}}) = n_i e \sqrt{\frac{eT_e}{m_i}} \exp\left(-\frac{1}{2} \right). \tag{4.7}$$

4.4.2.2.2.5 Exponential parameters to plasma parameters Since we have four plasma parameters and three exponential parameters, we can't invert the entire fitting procedure. Luckily, the shells can very accurately fit the plasma potential and since the plasma potential for both the shell and core should be the same we use that as a constant input. Thus we have

$$n_e(A, B) = A \Bigg/ e \sqrt{\frac{eB}{m_i}} \left[\frac{1}{2} \sqrt{\frac{2m_i}{\pi m_e}} \exp\left(\frac{-V_{\text{plasma}}}{B} \right) \right] \tag{4.8}$$

$$T_e(B) = B \tag{4.9}$$

$$n_i(C, B) = C \Bigg/ e \sqrt{\frac{eB}{m_i}} \exp\left(-\frac{1}{2} \right). \tag{4.10}$$

4.4.2.2.2.6 Converting fit errors After fitting the exponential parameters, our fitting routine spits out a set of standard deviations for each fit parameter. Assuming linear error then we can write standard deviations on each of the plasma parameters as

$$\sigma_{ne} = \partial_A n_e(A, B)\sigma_A + \partial_B n_e(A, B)\sigma_B \tag{4.11}$$

$$\sigma_{Te} = \sigma_B \tag{4.12}$$

$$\sigma_{ni} = \partial_C n_e(C, B)\sigma_C + \partial_B n_e(C, B)\sigma_B. \tag{4.13}$$

4.4.2.3 Extracting plasma parameters

Using any of the aforementioned theories we can get individual plasma parameters for each tip of the PA probe but now we need to get the mean plasma parameters for all of the tips. It's simple enough to get the mean density and temperature of the tips by just taking an average but use spherical harmonics to better extract the parameters of interest. We've also chosen to ignore evaluation of the drift velocity from the ion densities because of the increased complexity and the drift velocity is not as much of an interest as the temperature anisotropy.

4.4.2.3.1 Spherical harmonics Since each tip is measuring it's own value, we need some way to extract the temperature anisotropy. A good way to do that is to take spherical harmonics of the system with the magnetic field direction being the polar axis. By using spherical harmonics that only vary as a function of polar angle, θ, this let's get only the components either parallel or perpendicular to the magnetic field. To note, the distribution or measurements may not be symmetric about the magnetic field but we only care for the pressure anisotropy which is gyroaveraged.

4.4.2.3.1.1 Temperature by direction If we ignore the drift velocity in the distribution function, equation 4.2, we find that the term in the exponential coefficient takes the form

$$-\frac{em_e}{2}v^2\left(\frac{\cos^2\theta}{T_\parallel} + \frac{\sin^2\theta}{T_\perp}\right) \tag{4.14}$$

where θ is the angle between the magnetic field and the velocity (whose magnitude is given by v). Thus we can write a directional temperature as

$$T(\theta) = \frac{T_\perp T_\parallel}{T_\parallel \sin^2\theta + T_\perp \cos^2\theta}. \tag{4.15}$$

This can not be written in a finite number of spherical harmonics but the inverse temperature can:

$$\begin{aligned}
\frac{1}{T(\theta)} &= \frac{T_\parallel \sin^2\theta + T_\perp \cos^2\theta}{T_\perp T_\parallel} && (4.16)\\
&= \frac{T_\parallel}{T_\perp T_\parallel} + \frac{T_\perp - T_\parallel}{T_\perp T_\parallel}\cos^2\theta && (4.17)\\
&= \frac{2\sqrt{\pi}}{3}\left(\frac{2T_\parallel + T_\perp}{T_\parallel T_\perp} Y_{0,0}(\theta,\phi) - \frac{2}{\sqrt{5}}\frac{T_\parallel - T_\perp}{T_\parallel T_\perp} Y_{2,0}(\theta,\phi)\right) && (4.18)
\end{aligned}$$

where $Y_{i,j}$ are the spherical harmonics.

Now the mean temperature of all the tips is not necessarily the mean temperature gotten from the second moment of the distribution function. From the distribution function, the mean temperature is

$$\frac{1}{2} m_e \iiint v^2 f(v)\, d\mathbf{v} = \frac{T_\parallel + 2T_\perp}{3}. \tag{4.19}$$

If we assume that each tip only measures the distribution function along a ray in the direction of the tip then we need to average the directional temperature, equation 4.15. As we are really fitting $1/T$ we find that

$$T_{avg} = \left(\frac{\iint 1/T(\theta)\, d\Omega}{\iint d\Omega}\right)^{-1} = \frac{3T_\parallel T_\perp}{2T_\parallel + T_\perp}. \tag{4.20}$$

To note, if we were to instead average the temperature, not the inverse

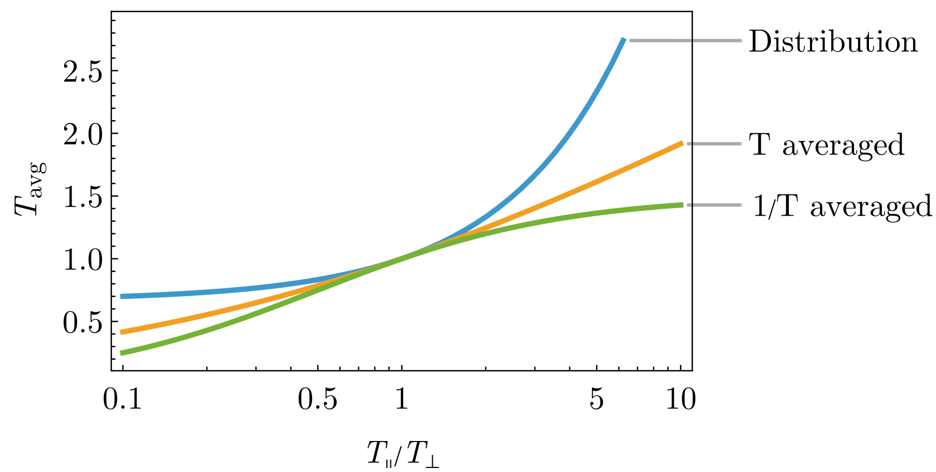


Figure 4.20: A comparison of the various methods for averaging the temperature. Averaging the distribution always leads to the largest average temperature whereas averaging the temperature by direction scales at a different rate.

temperature, we would find

$$\mathrm{T}_{\text{avg,non-inverse}} = \begin{cases} \frac{\mathrm{T}_{\parallel}\mathrm{T}_{\perp}}{\sqrt{\mathrm{T}_{\parallel}\left(\mathrm{T}_{\perp}-\mathrm{T}_{\parallel}\right)}}\cos^{-1}\left(\sqrt{\frac{\mathrm{T}_{\parallel}}{\mathrm{T}_{\perp}}}\right) & \text{if } \mathrm{T}_{\parallel} < \mathrm{T}_{\perp} \\ \frac{\mathrm{T}_{\parallel}\mathrm{T}_{\perp}}{\sqrt{\mathrm{T}_{\parallel}\left(\mathrm{T}_{\parallel}-\mathrm{T}_{\perp}\right)}}\cosh^{-1}\left(\sqrt{\frac{\mathrm{T}_{\parallel}}{\mathrm{T}_{\perp}}}\right) & \text{if } \mathrm{T}_{\parallel} > \mathrm{T}_{\perp} \end{cases} \tag{4.21}$$

which is larger than T_{avg} for all $\mathrm{T}_{\parallel}/\mathrm{T}_{\perp}$. This can be seen in figure 4.20.

4.4.2.3.2 Error bars Monte-carlo

4.5 Curve Fitting

We will use ODR like mentioned in Te section.

4.5.1 Combining multiple shots

We want to build up a full IV-characteristic so that we can fit the various plasma potentials. To do so, we take shots at a variety of different bias voltages and with the core current paths in both orientations, see 4.3.2.4. In total, this requires on the order of 100 shots so we need to make sure that these shots are sufficiently similar in plasma dynamics and the time coordinate is correctly aligned.

4.5.1.1 Timing alignment

Since each shot is slightly different (although quite reproducable), we need to adjust the time index for each shot such that the features in the shot match up nicely for all the shots. An example of this can be seen in Fig. 4.21 where the actual shot timing is eight time samples (0.8 μs) after the normal shot timing. We can correct this in one of two ways:

4.5.1.1.1 B-dot cross-correlation Cross-correlation is a method of finding when different signals align by applying different amounts of offset between the two signals and calculating

$$C_{i,j}(\Delta t) = \frac{1}{T_i T_j} \int \mathbf{S}_i(t) \cdot \mathbf{S}_j(t + \Delta t)\, dt. \tag{4.22}$$

Both of our signals are of the same length but because we are shifting them by Δt the amount of time they overlap goes as $T_i = T - \Delta t$. Additionally as we have a discrete set of sample points, this is a sum in our case and subtracting off the mean of both signals before multiplying them (i.e. $\mathbf{S} \to \mathbf{S} - \overline{S}$) helps with correlation and is common place.

If we take the cross-correlation of the probe B-dot signals in machine coordinates we can maximize that to find when the signals for that shot best align with the average shot.

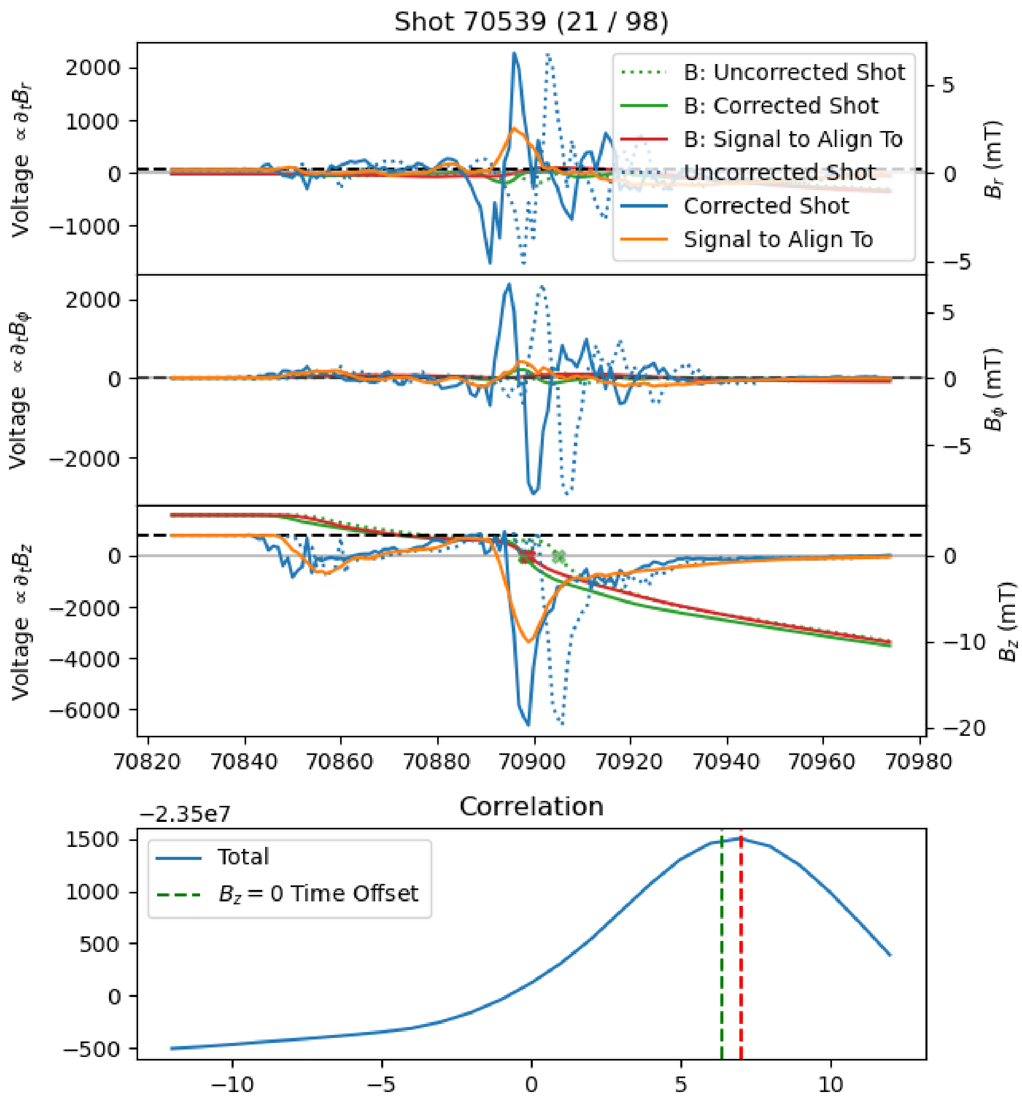


Figure 4.21: The alignment procedure for a single shot. Alignment can occur in one of two ways. Either the cross-correlation of the B-dot signals can be maximized or the crossing time of $B_z = 0$ can be aligned. Here we've chosen a particularly egregious shot for alignment purposes since the alignment wants to offset the arrays by 8 time samples for the B-dot cross-correlation (0.8 μs) or 7 time samples for the $B_z = 0$ method (red and green dashed lines respectively in bottom panel). The top three panels show the unaligned data from the shot (blue dotted for both B-dot and B). The orange and red lines are the data to align with which is found using the mean of the B-dot and B signals for all of the shots. The Xs in the $\partial_t B_z$ panel are where $B_z = 0$ for the unaligned and aligned data.

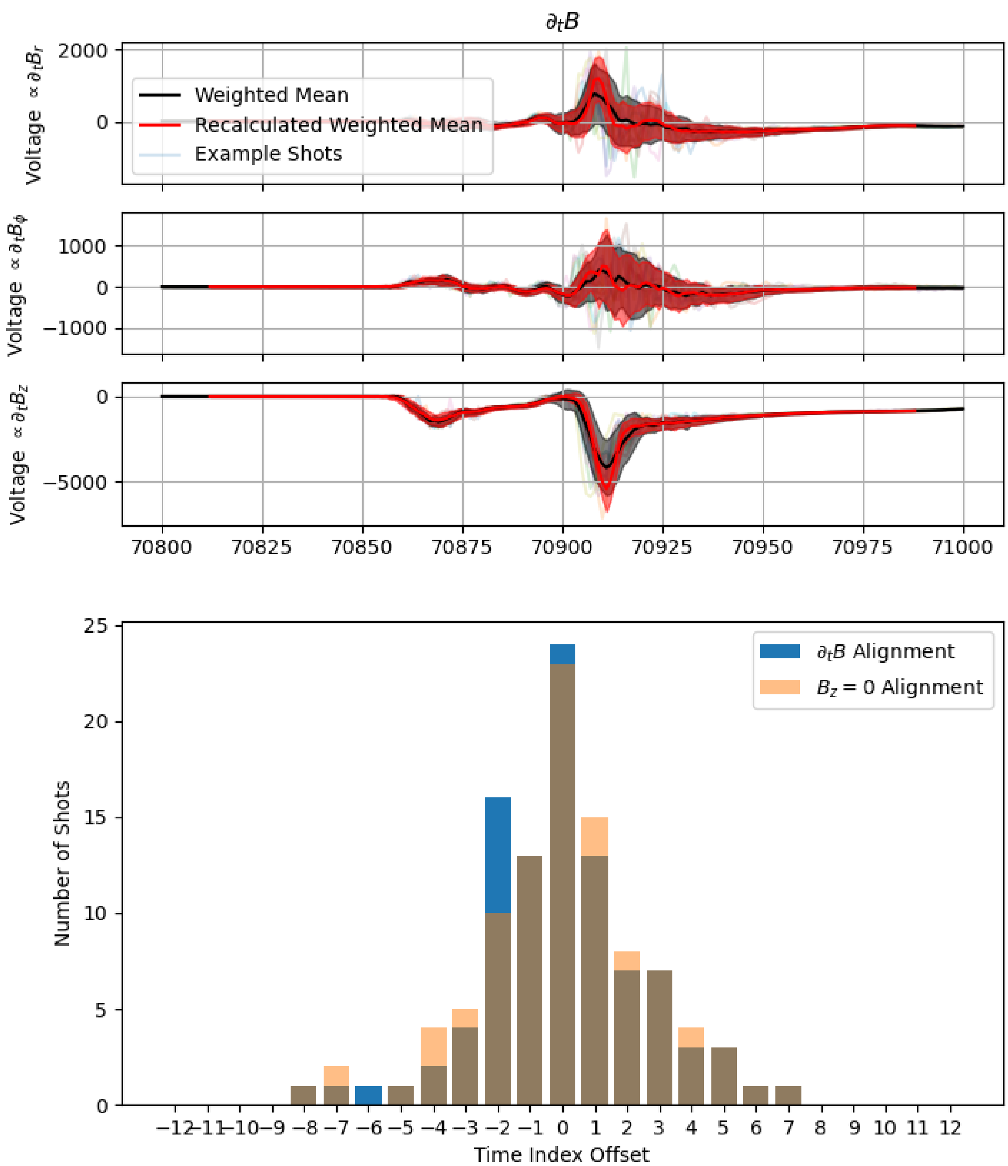


Figure 4.22: An example of the effects of aligning data for the PA probe. Using 98 shots we have aligned the data in time using either cross-correlation of the B-dot signals on the probe or by aligning the $B_z = 0$ crossing time index. The differences in the alignment method are seen in the bottom figure. The histograms for both methods are similar and thus give similar alignment results. The top three panels show example B-dot traces (light colored lines). The average of all 98 signals before alignment is seen in black with one standard deviation bounds. After alignment, a new mean is calculated and shown in red. It can be seen that the standard deviation is far smaller after alignment and the red mean has less smoothed features.

4.5.1.1.1.1 B-dot smoothing During the initial phase when the capacitor bank trigger activates and later when the bank switch turns on, there is an enormous amount of ringing in the lab which can lead to oscillations at the start of the signal. Since the trigger is very uniform between shots, this can lead to alignment at the trigger time, not alignment of the plasma features. Thus we smooth out the start of the shot so that the ringing is removed.

4.5.1.1.2 Bz = 0 alignment Instead of aligning the entire B-dot signal, we can instead focus only on when the reconnection layer passes and the reconnecting magnetic field, B_z, reverses direction.

4.5.1.1.2.1 Aligns where we expect Anisotropy This has the advantage of aligning the plasma features in the outflow of reconnection which is where we are searching for signs of pressure anisotropy (or lack thereof).

4.5.1.1.2.2 Depends heavily on prior bdot data Unfortunately, since we calculate **B** by integrating the B-dot signal this depends on the signals measured earlier in time and because these coils aren't optimized for the high frequency measurement we are doing this can lead to incorrect integrated fields because of ringing in the B-dot circuit or non-linear transfer functions.

4.5.1.1.3 Delay cause This delay in firing that both alignment methods attempt to account for is due to the capacitor bank thyratron (for capacitor bank layout see Endrizzi (2021)). The trigger, which is a spark in the gas inside the thyratron, can lead to a variety of different full conduction times so there is a distribution of drive start times which can be seen in the bottom panel of Fig. 4.22. Additionally, the reason for different reconnection timings may also be due to slight variations in how the layer forms and reconnects.

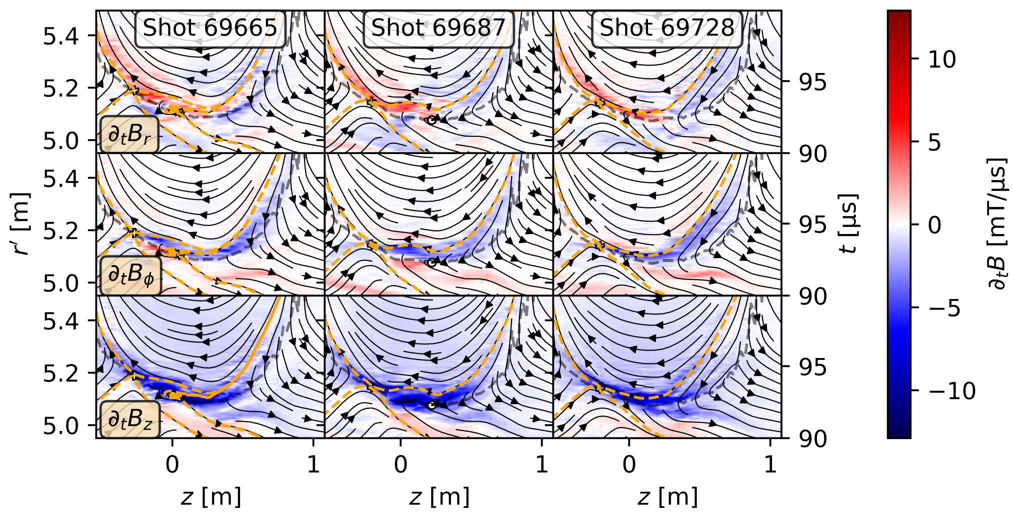


Figure 4.23: This figure shows three different shots (columns) and their respective Bdot measurements in various directions (rows). Most figures are like that of shot 69728 where there is a single x-point for many of our configurations but sometimes during a shot, plasmoids can form like those in shots 69664 and 69687. These plasmoids can form either to the left or right of the main reconnecting x-point and can change the outflow dynamics of the reconnecting current layer.

4.5.1.1.4 Result comparison It is difficult to quantify which method works better for alignment because of a lack in ground truth knowledge, the method behind B-dot cross-correlation appears more sound and is less susceptable to noise which is widespread in the lab. Both methods lead to very similar results which can be seen in the bottom panel of 4.22 where both histograms are very similar in shape.

4.5.1.2 Shot selection

We have to use many shots to build up a full IV-characteristic for the PA probe and different machine setups/parameters lead to different amounts of reproducibility. In the plasmas we are studying while looking for pressure anisotropy, we find the dominant source of irreproducibility are plasmoids that form near the x-point and are ejected out (shots 69665 and 69687 in figure 4.23). It is relatively easy to pick out shots that vary drastically by eye from these lightsaber probe measurements (see 1.3.5.1.2.3 for the probe description).

We can also remove bad shots by comparing all shots where the PA probe was at the same bias. If a shot gives signals for all the tips that are very different than the other shots we remove the outlier shot.

4.5.1.3 Averaging of core data

To average the data we start by finding clusters of shots where the bias voltage, V_{bias}, is within some tolerance of each other (usually 5 V). Then we calculate I_{probe} and V_{probe} for each shot. For each shot cluster of size N there will be n shots with the relay off and $N-n$ shots with the relay on. Then our averaged data can be written as

$$\begin{aligned} I^{\text{avg}}_{\text{probe}} &= \frac{1}{2}\left(\frac{1}{n}\sum_{i\in n} I^{i}_{\text{probe}} + \frac{1}{N-n}\sum_{i\in N-n} I^{i}_{\text{probe}}\right) \\ &= \frac{1}{2n(N-n)}\sum_{i\in n}\sum_{j\in N-n}\left(I^{i}_{\text{probe}} + I^{j}_{\text{probe}}\right) \\ V^{\text{avg}}_{\text{probe}} &= \frac{1}{2n(N-n)}\sum_{i\in n}\sum_{j\in N-n}\left(V^{i}_{\text{probe}} + V^{j}_{\text{probe}}\right) \end{aligned}$$

Here we have normalized our averaging terms by the number of shots in each cluster.

4.5.2 Fitting shell

4.5.2.1 Use all shell data

4.5.2.2 Fit using BRL

4.5.2.3 Fits pretty good

4.5.3 Fitting core

4.5.3.1 Fitting using full routine

4.5.3.1.1 Slow

4.5.3.1.2 Model not perfectly representative

4.5.3.2 Fitting tips independently

4.5.3.2.1 Individual fitting method

4.5.3.2.2 Holding Vprobe constant

4.5.3.2.3 Exponential fitting method

4.5.3.2.4 Spherical harmonics

4.6 Results

Figure out what my results are once I have some reasonable results

4.6.1 Drive cylinder

4.6.1.1 What did we try

4.6.1.2 Why didn't we see pressure anisotropy

4.6.1.2.1 In the bdot data

4.6.1.2.2 In the PA data

4.6.2 3 coil

4.6.2.1 How much anisotropy should we expect

4.6.2.2 What setups did we try

4.6.2.3 Old results of just comparing radar plots

4.6.2.4 New results using independent fitting

5 CONCLUSION

5.1 Summary of thesis

5.1.1 Using pressure waves to measure density

5.1.2 Upgrading langmuir probe analysis and new possibilities

5.1.3 Using directional langmuir probes for measuring pressure anisotropy

5.2 Future work

5.2.1 Pressure wave

5.2.1.1 Aforementioned future work

5.2.1.2 Extracting cross-field diffusion

5.2.2 Langmuir Probe

5.2.2.1 Prepared to analyze lots of cool data

5.2.2.2 Extract out electric field

5.2.3 PA Probe

5.2.3.1 Try more configurations

5.2.3.2 Map out full pressure anisotropy during shot

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This template uses Gyre Pagella by default. (I used Arno Pro in my dissertation.)

Feel free to give me a shout-out in your colophon or acks if this template is useful for you. Good luck!

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